Chapter 3

Measurement System Behavior

3.1 INTRODUCTION

This chapter introduces the concept of simulating measurement system behavior through mathematical modeling. From such a modeling study, the important aspects of measurement system response that are pertinent to system design and specification can be obtained. Each measurement system responds differently to different types of input signals and to the dynamic content within these signals. So a particular system may not be suitable for measuring certain signals or at least portions of some signals. Yet a measurement system always provides information regardless of how well (or poorly!) this reflects the actual input signal being measured. To explore this concept, this chapter discusses system response to certain types of input signals.

Throughout this chapter, we use the term ‘‘measurement system’’ in a generic sense. It refers either to the response of the measurement system as a whole or to the response of any component or instrument that makes up that system. Either is important, and both are interpreted in similar ways. Each individual stage of the measurement system has its own response to a given input. The overall system response is affected by the response of each stage of the complete system.

Upon completion of this chapter, the reader will be able to

- relate generalized measurement system models to dynamic response,
- describe and analyze models of zero-, first-, and second-order measurement systems and predict their general behavior,
- calculate static sensitivity, magnitude ratio, and phase shift for a range of systems and input waveforms,
- state the importance of phase linearity in reducing signal distortion,
- analyze the response of a measurement system to a complex input waveform, and
- determine the response of coupled measurement systems.

3.2 GENERAL MODEL FOR A MEASUREMENT SYSTEM

As pointed out in Chapter 2, all input and output signals can be broadly classified as being static, dynamic, or some combination of the two. For a static signal, only the signal magnitude is needed to reconstruct the input signal based on the indicated output signal. Consider measuring the length of a board using a ruler. Once the ruler is positioned, the indication (output) of the magnitude of length is immediately displayed because the board length (input) does not change over the time required to make the measurement. Thus, the board length represents a static input signal that is interpreted through the static magnitude output indicated by the ruler. But consider measuring the vibration of a motor. Vibration signals vary in amplitude and time, and thus are a dynamic input signal to the measuring instrument. But the ruler is not very useful in determining this dynamic information, so we need an instrument that can follow the input time signal faithfully.

Dynamic Measurements

For dynamic signals, signal amplitude, frequency, and general waveform information is needed to reconstruct the input signal. Because dynamic signals vary with time, the measurement system must be able to respond fast enough to keep up with the input signal. Further, we need to understand how the input signal is applied to the sensor because that plays a role in system response. Consider the time response of a common bulb thermometer for measuring body temperature. The thermometer, initially at approximately room temperature, is placed under the tongue. But even after several seconds, the thermometer does not indicate the expected value of body temperature and its display continually changes. What has happened? Surely your body temperature is not changing. If you were to use the magnitude of the output signal after only several seconds, you would come to a false conclusion about your health! Experience shows that within a few minutes, the correct body temperature will be indicated; so we wait. Experience also tells us that if we need accurate information faster, we would need a different type of temperature sensor. In this example, body temperature itself is constant (static) during the measurement, but the input signal to the thermometer is suddenly changed from room temperature to body temperature, that is, mathematically, a step change. This is a dynamic event as the thermometer (the measurement system) sees it! The thermometer must gain energy from its new environment to reach thermal equilibrium, and this takes a finite amount of time. The ability of any measurement system to follow dynamic signals is a characteristic of the design of the measuring system components.

Now consider the task of assessing the ride quality of an automobile suspension system. A simplified view for one wheel of this system is shown in Figure 3.1. As a tire moves along the road, the road surface provides the time-dependent input signal, $F(t)$, to the suspension at the tire contact

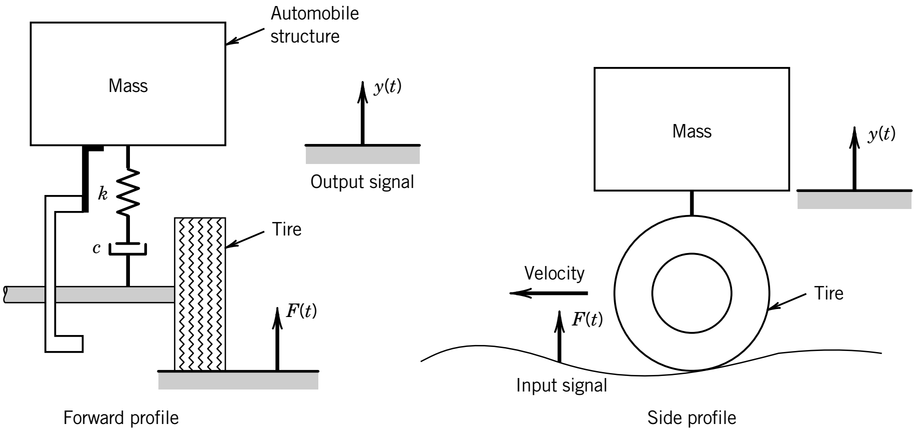


Figure 3.1 Lumped parameter model of an automobile suspension showing input and output signals.

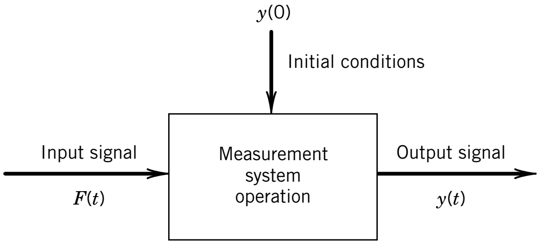


Figure 3.2 Measurement system operation on an input signal, $F(t)$, provides the output signal, $y(t)$.

point. The motion sensed by the passengers, $y(t)$, is a basis for the ride quality and can be described by a waveform that depends on the input from the road and the behavior of the suspension. An engineer must anticipate the form of the input signals so as to design the suspension to attain a desirable output signal.

Measurement systems play a key role in documenting ride quality. But just as the road and car interact to provide ride quality, the input signal and the measurement system interact in creating the output signal. In many situations, the goal of the measurement is to deduce the input signal based on the output signal. Either way, we see that it is important to understand how a measurement system responds to different forms of input signals.

The general behavior of measurement systems for a few common inputs defines, for the most part, the input–output signal relationships necessary to correctly interpret measured signals. We will show that only a few measurement system characteristics (specifications) are needed to predict the system response.

With the previous discussion in mind, consider that the primary task of a measurement system is to sense an input signal and to translate that information into a readily understandable and quantifiable output form. We can reason that a measurement system performs some mathematical operation on a sensed input. In fact, a general measurement system can be represented by a differential equation that describes the operation that a measurement system performs on the input signal. This concept is illustrated in Figure 3.2. For an input signal, $F(t)$, the system performs some operation that yields the output signal, $y(t)$. Then we must use $y(t)$ to infer $F(t)$. Therefore, at least a qualitative understanding of the operation that the measurement system performs is imperative to correctly interpret the input signal. We will propose a general mathematical model for a measurement system. Then, by representing a typical input signal as some function that acts as an input to the model, we can study just how the measurement system would behave by solving the model equation. In essence, we perform the analytical equivalent of a system calibration. This information can then be used to determine those input signals for which a particular measurement system is best suited.

Measurement System Model

In this section, we apply lumped parameter modeling to measurement systems. In lumped parameter modeling, the spatially distributed physical attributes of a system are modeled as discrete elements. The automotive suspension model discussed earlier is a lumped parameter model. As a simpler example, the mass, stiffness, and damping of a coil spring are properties spatially distributed along its length, but these can be replaced by the discrete elements of a mass, spring, and damper. An advantage is that the governing equations of the models reduce from partial to ordinary differential equations.

Consider the following general model of a measurement system, which consists of an nth-order linear ordinary differential equation in terms of a general output signal, represented by variable $y(t)$, and subject to a general input signal, represented by the forcing function, $F(t)$:

$$a_n \frac{d^n y}{dt^n} + a_{n-1} \frac{d^{n-1} y}{dt^{n-1}} + \cdots + a_1 \frac{dy}{dt} + a_0 y = F(t) \tag{3.1}$$

where

$$F(t) = b_m \frac{d^m x}{dt^m} + b_{m-1} \frac{d^{m-1} x}{dt^{m-1}} + \cdots + b_1 \frac{dx}{dt} + b_0 x \quad m \le n$$

The coefficients $a_0, a_1, a_2, \ldots, a_n$ and $b_0, b_1, \ldots, b_m$ represent physical system parameters whose properties and values depend on the measurement system itself. Real measurement systems can be modeled this way by considering their governing system equations. These equations are generated by application of pertinent fundamental physical laws of nature to the measurement system. Our discussion is limited to measurement system concepts, but a general treatment of systems can be found in text books dedicated to that topic (1–3).

Example 3.1

As an illustration, consider the seismic accelerometer depicted in Figure 3.3a. Various configurations of this instrument are used in seismic and vibration engineering to determine the motion of large bodies to which the accelerometer is attached. Basically, as the small accelerometer mass reacts to motion, it places the piezoelectric crystal into compression or tension, causing a surface charge to develop on the crystal. The charge is proportional to the motion. As the large body moves, the mass of the accelerometer will move with an inertial response. The stiffness of the spring, k, provides a restoring force to move the accelerometer mass back to equilibrium while internal frictional damping, c, opposes any displacement away from equilibrium. A model of this

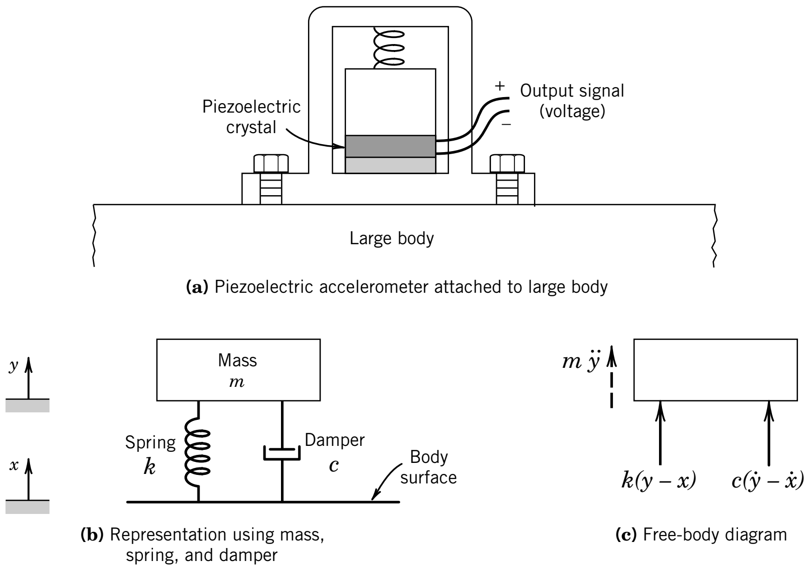


Figure 3.3 Lumped parameter model of accelerometer (Ex. 3.1).

measurement device in terms of ideal lumped elements of stiffness, mass, and damping is given in Figure 3.3b and the corresponding free-body diagram in Figure 3.3c. Let y denote the position of the small mass within the accelerometer and x denote the displacement of the body. Solving Newton's second law for the free body yields the second-order linear, ordinary differential equation

$$m\frac{d^2y}{dt^2} + c\frac{dy}{dt} + ky = c\frac{dx}{dt} + kx$$

Since the displacement y is the pertinent output from the accelerometer due to displacement x, the equation has been written such that all output terms, that is, all the y terms, are on the left side. All other terms are to be considered as input signals and are placed on the right side. Comparing this to the general form for a second-order equation ($n = 2$; $m = 1$) from Equation 3.1,

$$a^2\frac{d^2y}{dt^2} + a_1\frac{dy}{dt} + a_0y = b_1\frac{dx}{dt} + b_0x$$

we can see that $a_2 = m$, $a_1 = b_1 = c$, $a_0 = b_0 = k$, and that the forces developed due to the velocity and displacement of the body become the inputs to the accelerometer. If we could anticipate the waveform of x, for example, $x(t) = x_0 \sin \omega t$, we could solve for $y(t)$, which gives the measurement system response.

Fortunately, many measurement systems can be modeled by zero-, first-, or second-order linear, ordinary differential equations. More complex systems can usually be simplified to these lower orders. Our intention here is to attempt to understand how systems behave and how such response is closely related to the design features of a measurement system; it is not to simulate the exact system behavior. The exact input–output relationship is found from calibration. But modeling guides us in choosing specific instruments and measuring methods by predicting system response to signals, and in determining the type, range, and specifics of calibration. Next, we examine several special cases of Equation 3.1 that model the most important concepts of measurement system behavior.

3.3 SPECIAL CASES OF THE GENERAL SYSTEM MODEL

Zero-Order Systems

The simplest model of a measurement systems and one used with static signals is the zero-order system model. This is represented by the zero-order differential equation:

$$a_0y = F(t)$$

Dividing through by a_0 gives

$$y(t) = KF(t) \tag{3.2}$$

where $K = 1/a_0$. K is called the static sensitivity or steady gain of the system. This system property was introduced in Chapter 1 as the relation between the change in output associated with a change in static input. In a zero-order model, the system output is considered to respond to the input signal instantaneously. If an input signal of magnitude $F(t) = A$ were applied, the instrument would indicate KA, as modeled by Equation 3.2. The scale of the measuring device would be calibrated to indicate A directly.

For real systems, the zero-order system concept is used to model the non–time-dependent measurement system response to static inputs. In fact, the zero-order concept appropriately models any system during a static calibration. When dynamic input signals are involved, a zero-order model is valid only at static equilibrium. This is because most real measurement systems possess inertial or storage capabilities that require higher-order differential equations to correctly model their time-dependent behavior to dynamic input signals.

Determination of *K*

The static sensitivity is found from the static calibration of the measurement system. It is the slope of the calibration curve, $K = dy/dx$.

Example 3.2

A pencil-type pressure gauge commonly used to measure tire pressure can be modeled at static equilibrium by considering the force balance on the gauge sensor, a piston that slides up and down a cylinder. An exploded view of an actual gauge is shown in Figure 3.4c. In Figure 3.4a, we model the piston motion[1] as being restrained by an internal spring of stiffness, k, so that at static equilibrium the absolute pressure force, F, bearing on the piston equals the force exerted on the piston by the spring, F_s, plus the atmospheric pressure force, F_{atm}. In this manner, the transduction of pressure into mechanical displacement occurs. Considering the piston free-body at static equilibrium in Figure 3.4b, the static force balance, $\Sigma F = 0$, gives

$$ky = F - F_{atm}$$

where y is measured relative to some static reference position marked as zero on the output display. Pressure is simply the force acting inward over the piston surface area, A. Dividing through by area

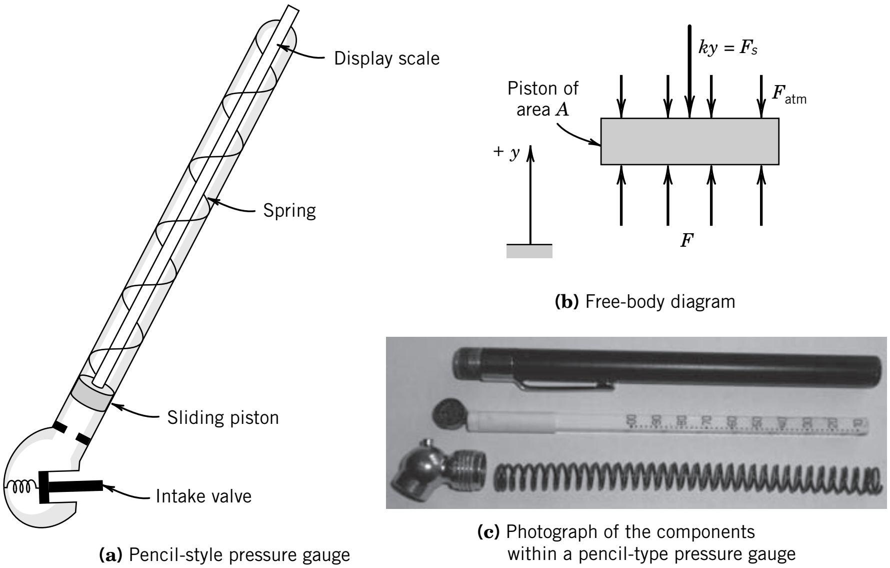


Figure 3.4 Lumped parameter model of pressure gauge (Ex. 3.2).

[1] In a common gauge there may or may not be the mechanical spring shown.

provides the zero-order response equation between output displacement and input pressure and gives

$$y = (A/k)(p - p_{\text{atm}})$$

The term $(p - p_{\text{atm}})$ represents the pressure relative to atmospheric pressure. It is the pressure indicated by this gauge. Direct comparison with Equation 3.2 shows that the input pressure magnitude equates to the piston displacement through the static sensitivity, $K = A/k$. Drawing from the concept of Figure 3.2, the system operates on pressure so as to bring about the relative displacement of the piston, the magnitude of which is used to indicate the pressure. The equivalent of spring stiffness and piston area affect the magnitude of this displacement—factors considered in its design. The exact static input–output relationship is found through calibration of the gauge. Because elements such as piston inertia and frictional dissipation were not considered, this model would not be appropriate for studying the dynamic response of the gauge.

First-Order Systems

Measurement systems that contain storage elements do not respond instantaneously to changes in input. The bulb thermometer discussed in Section 3.2 is a good example. The bulb exchanges energy with its environment until the two are at the same temperature, storing energy during the exchange. The temperature of the bulb sensor changes with time until this equilibrium is reached, which accounts physically for its lag in response. The rate at which temperature changes with time can be modeled with a first-order derivative and the thermometer behavior modeled as a first-order equation. In general, systems with a storage or dissipative capability but negligible inertial forces may be modeled using a first-order differential equation of the form

$$a_1\dot{y} + a_0 y = F(t) \tag{3.3}$$

with $\dot{y} = dy/dt$. Dividing through by a_0 gives

$$\tau y + y = KF(t) \tag{3.4}$$

where $\tau = a_1/a_0$. The parameter τ is called the *time constant* of the system. Regardless of the physical dimensions of a_0 and a_1, their ratio will always have the dimensions of time. The time constant provides a measure of the speed of system response, and as such is an important specification in measuring dynamic input signals. To explore this concept more fully, consider the response of the general first-order system to the following two forms of an input signal: the step function and the simple periodic function.

Step Function Input

The step function, $AU(t)$, is defined as

$$AU(t) = 0 \quad t \le 0^-$$
$$AU(t) = A \quad t \ge 0^+$$

where A is the amplitude of the step function and $U(t)$ is defined as the unit step function as depicted in Figure 3.5. Physically, this function describes a sudden change in the input signal from a constant value of one magnitude to a constant value of some other magnitude, such as a sudden change in

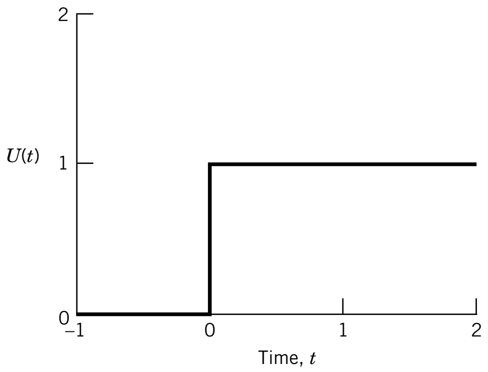


Figure 3.5 The unit step function, $U(t)$.

loading, displacement, or any physical variable. When we apply a step function input to a measurement system, we obtain information about how quickly a system will respond to a change in input signal. To illustrate this, let us apply a step function as an input to the general first-order system. Setting $F(t) = AU(t)$ in Equation 3.4 gives

$$\tau\dot{y} + y = KAU(t) = KF(t)$$

with an arbitrary initial condition denoted by, $y(0) = y_0$. Solving for $t \geq 0^+$ yields

$$\underbrace{y(t)}_{\text{Time response}} = \underbrace{KA}_{\text{Steady response}} + \underbrace{(y_0 - KA)e^{-t/\tau}}_{\text{Transient response}} \tag{3.5}$$

The solution of the differential equation, $y(t)$, is the time response (or simply the response) of the system. Equation 3.5 describes the behavior of the system to a step change in input. This means that $y(t)$ is in fact the output indicated by the display stage of the system. It should represent the time variation of the output display of the measurement system if an actual step change were applied to the system. We have simply used mathematics to simulate this response.

Equation 3.5 consists of two parts. The first term is known as the steady response because, as $t \to \infty$, the response of $y(t)$ approaches this steady value. The steady response is that portion of the output signal that remains after the transient response has decayed to zero. The second term on the right side of Equation 3.5 is known as the transient response of $y(t)$ because, as $t \to \infty$, the magnitude of this term eventually reduces to zero.

For illustrative purposes, let $y_0 < A$ so that the time response becomes as shown in Figure 3.6. Over time, the indicated output value rises from its initial value, at the instant the change in input is applied, to an eventual constant value, $y_\infty = KA$, at steady response. As an example, compare this general time response to the recognized behavior of the bulb thermometer when measuring body temperature as discussed earlier. We see a qualitative similarity. In fact, in using a bulb thermometer to measure body temperature, this is a real step function input to the thermometer, itself a first-order measuring system.

Suppose we rewrite the response Equation 3.5 in the form

$$\Gamma(t) = \frac{y(t) - y_\infty}{y_0 - y_\infty} = e^{-t/\tau} \tag{3.6}$$

The term $\Gamma(t)$ is called the *error fraction* of the output signal. Equation 3.6 is plotted in Figure 3.7, where the time axis is nondimensionalized by the time constant. We see that the error fraction

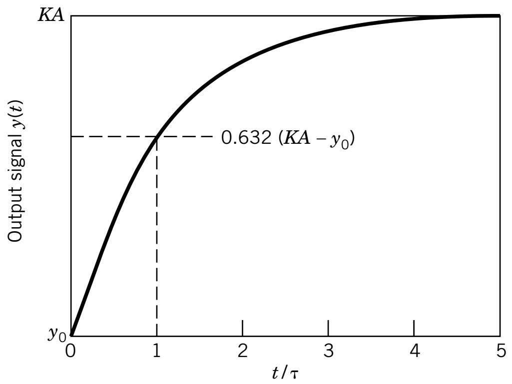


Figure 3.6 First-order system time response to a step function input: the time response, $y(t)$.

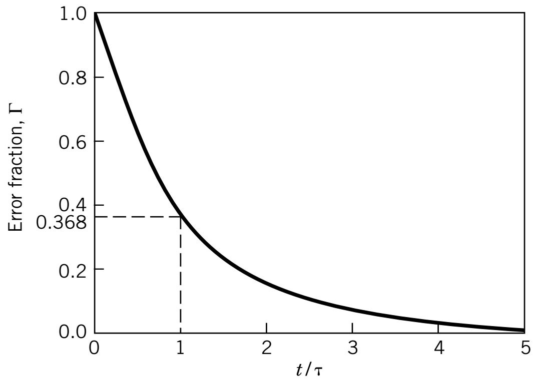


Figure 3.7 First-order system time response to a step function input: the error fraction, Γ.

decreases from a value of 1 and approaches a value of 0 with increasing t/τ. At the instant just after the step change in input is introduced, $\Gamma = 1.0$, so that the indicated output from the measurement system is 100% in error. This implies that the system has responded 0% to the input change, that is, it has not changed. From Figure 3.7, it is apparent that as time moves forward the system does respond, and with decreasing error in its indicated value. Let the percent response of the system to a step change be given as $(1 - \Gamma) \times 100$. Then by $t = \tau$, where $\Gamma = 0.368$, the system will have responded to 63.2% of the step change. Further, when $t = 2.3\tau$ the system will have responded $(\Gamma = 0.10)$ to 90% of the step change; by $t = 5\tau$, we find the response to be 99.3%. Values for the percent response and the corresponding error as functions of t/τ are summarized in Table 3.1. The time required for a system to respond to a value that is 90% of the step input, $y_\infty - y_0$, is important and is called the *rise time* of the system.

Based on this behavior, we can see that the time constant is in fact a measure of how quickly a first-order measurement system will respond to a change in input value. A smaller time constant indicates a shorter time between the instant that an input is applied and when the system reaches an essentially steady output. We define the *time constant* as the time required for a first-order system to achieve 63.2% of the step change magnitude, $y_\infty - y_0$. The time constant is a system property.

Determination of τ From the development above, the time constant can be experimentally determined by recording the system's response to a step function input of a known magnitude.

Table 3.1 First-Order System Response and Error Fraction

t/τ	% Response	Γ	% Error
0	0.0	1.0	100.0
1	63.2	0.368	36.8
2	86.5	0.135	13.5
2.3	90.0	0.100	10.0
3	95.0	0.050	5.0
5	99.3	0.007	0.7
∞	100.0	0.0	0.0

In practice, it is best to record that response from $t = 0$ until steady response is achieved. The data can then be plotted as error fraction versus time on a semilog plot, such as in Figure 3.8. This type of plot is equivalent to the transformation

$$\ln\Gamma = 2.3\log\Gamma = -(1/\tau)t \tag{3.7}$$

which is of the linear form, $Y = mX + B$ (where $Y = \ln\Gamma$, $m = -(1/\tau)$, $X = t$, and $B = 0$ here). A linear curve fit through the data will provide a good estimate of the slope, m, of the resulting plot. From Equation 3.7, we see that $m = -1/\tau$, which yields the estimate for τ.

This method offers advantages over attempting to compute τ directly from the time required to achieve 63.2% of the step-change magnitude. First, real systems will deviate somewhat from perfect first-order behavior. On a semilog plot (Figure 3.8), such deviations are readily apparent as clear trends away from a straight line. Modest deviations do not pose a problem. But strong deviations indicate that the system is not behaving as expected, thus requiring a closer examination of the system operation, the step function experiment, or the assumed form of the system model. Second, acquiring data during the step function experiment is prone to some random error in each data point. The use of a data curve fit to determine τ utilizes all of the data over time so as to minimize the influence of an error in any one data point. Third, the method eliminates the need to determine the $\Gamma = 1.0$ and 0.368 points, which are difficult to establish in practice and so are prone to a systematic error.

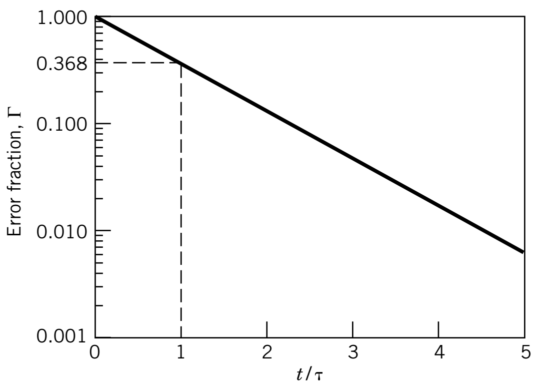


Figure 3.8 The error fraction plotted on semilog coordinates.

Example 3.3

Suppose a bulb thermometer originally indicating 20°C is suddenly exposed to a fluid temperature of 37°C. Develop a model that simulates the thermometer output response.

KNOWN $T(0) = 20°\text{C}$

$T_\infty = 37°\text{C}$

$F(t) = [T_\infty - T(0)]U(t)$

ASSUMPTIONS To keep things simple, assume the following: no installation effects (neglect conduction and radiation effects); sensor mass is mass of liquid in bulb only; uniform temperature within bulb (lumped mass); and thermometer scale is calibrated to indicate temperature

FIND $T(t)$

SOLUTION Consider the energy balance developed in Figure 3.9. According to the first law of thermodynamics, the rate at which energy is exchanged between the sensor and its environment through convection, $\dot{Q}$, must be balanced by the storage of energy within the thermometer, dE/dt. This conservation of energy is written as

$$\frac{dE}{dt} = \dot{Q}$$

Energy storage in the bulb is manifested by a change in bulb temperature so that for a constant mass bulb, $dE(t)/dt = mc_v dT(t)/dt$. Energy exchange by convection between the bulb at $T(t)$ and an environment at T_∞ has the form $\dot{Q} = hA_s \Delta T$. The first law can be written as

$$mc_v \frac{dT(t)}{dt} = hA_s[T_\infty - T(t)]$$

This equation can be written in the form

$$mc_v \frac{dT(t)}{dt} + hA_s[T(t) - T(0)] = hA_s F(t) = hA_s[T_\infty - T(0)]U(t)$$

with initial condition $T(0)$ and

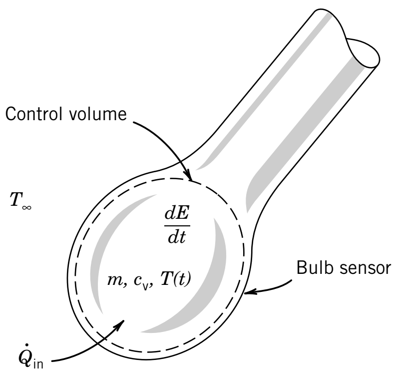


Figure 3.9 Lumped parameter model of thermometer and its energy balance (Ex. 3.3).

where

m = mass of liquid within thermometer
c_v = specific heat of liquid within thermometer
h = convection heat transfer coefficient between bulb and environment
A_s = thermometer surface area

The term hA_s controls the rate at which energy can be transferred between a fluid and a body; it is analogous to electrical conductance. By comparison with Equation 3.3, $a_0 = hA_s$, $a_1 = mc_v$, and $b_0 = hA_s$. Rewriting for $t \geq 0^+$ and simplifying yields

$$\frac{mc_v}{hA_s}\frac{dT(t)}{dt} + T(t) = T_\infty$$

From Equation 3.4, this implies that the time constant and static sensitivity are

$$\tau = \frac{mc_v}{hA_s} \quad K = \frac{hA_s}{hA_s} = 1$$

Direct comparison with Equation 3.5 yields this thermometer response:

$$\begin{aligned} T(t) &= T_\infty + [T(0) - T_\infty]e^{-t/\tau} \\ &= 37 - 17e^{-t/\tau} \ [^\circ C] \end{aligned}$$

COMMENT Two interactive examples of the thermometer problem (from Exs 3.3–3.5) are available. In the program *FirstOrd*, the user can choose input functions and study the system response. In the LabView program *Temperature_response*, the user can apply interactively a step change in temperature and study the first-order system response of a thermal sensor.

Clearly the time constant, τ, of the thermometer can be reduced by decreasing its mass-to-area ratio or by increasing h (for example, increasing the fluid velocity around the sensor). Without modeling, such information could be ascertained only by trial and error, a time-consuming and costly method with no assurance of success. Also, it is significant that we found that the response of the temperature measurement system in this case depends on the environmental conditions of the measurement that control h, because the magnitude of h affects the magnitude of τ. If h is not controlled during response tests (i.e., if it is an extraneous variable), ambiguous results are possible. For example, the curve of Figure 3.8 will become nonlinear, or replications will not yield the same values for τ.

Review of this example should make it apparent that the results of a well-executed step calibration may not be indicative of an instrument's performance during a measurement if the measurement conditions differ from those existing during the step calibration.

Example 3.4

For the thermometer in Example 3.3 subjected to a step change in input, calculate the 90% rise time in terms of t/τ.

KNOWN Same as Example 3.3

ASSUMPTIONS Same as Example 3.3

FIND 90% response time in terms of t/τ

SOLUTION The percent response of the system is given by $(1 - \Gamma) \times 100$ with the error fraction, Γ, defined by Equation 3.6. From Equation 3.5, we note that at $t = \tau$, the thermometer will indicate $T(t) = 30.75°C$, which represents only 63.2% of the step change from 20° to 37°C. The 90% rise time represents the time required for Γ to drop to a value of 0.10. Then

$$\Gamma = 0.10 = e^{-t/\tau}$$

or $t/\tau = 2.3$.

COMMENT In general, a time equivalent to 2.3τ is required to achieve 90% of the applied step input for a first-order system.

Example 3.5

A particular thermometer is subjected to a step change, such as in Example 3.3, in an experimental exercise to determine its time constant. The temperature data are recorded with time and presented in Figure 3.10. Determine the time constant for this thermometer. In the experiment, the heat transfer coefficient, h, is estimated to be 6 W/m^2-°C from engineering handbook correlations.

KNOWN Data of Figure 3.10

$h = 6\ \mathrm{W/m^2\text{-}°C}$

ASSUMPTIONS First-order behavior using the model of Example 3.3, constant properties

FIND τ

SOLUTION According to Equation 3.7, the time constant should be the negative reciprocal of the slope of a line drawn through the data of Figure 3.10. Aside from the first few data points, the data appear to follow a linear trend, indicating a nearly first-order behavior and validating our model assumption. The data is fit to the first-order equation[2]

$$2.3 \log \Gamma = (-0.194)t + 0.00064$$

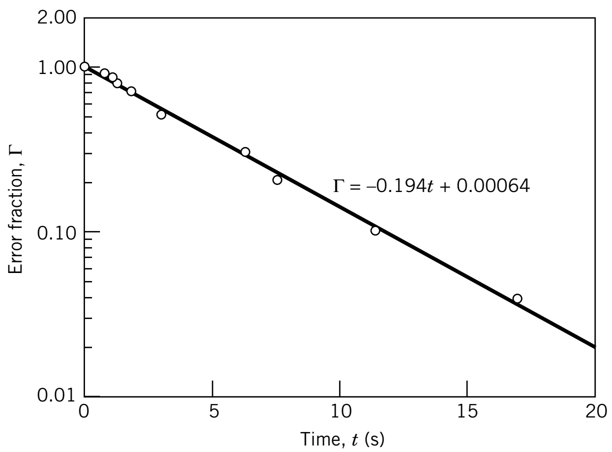


Figure 3.10 Temperature–time history of Example 3.5.

[2] The least-squares approach to curve fitting is discussed in detail in Chapter 4.

With $m = -0.194 = -1/\tau$, the time constant is calculated as $\tau = 5.15$ seconds.

COMMENT If the experimental data were to deviate significantly from first-order behavior, this would be a clue that either our assumptions do not fit the real-problem physics or that the test conduct has control or execution problems.

Simple Periodic Function Input

Periodic signals are commonly encountered in engineering processes. Examples include vibrating structures, vehicle suspension dynamics, biological circulations, and reciprocating pump flows. When periodic inputs are applied to a first-order system, the input signal frequency has an important influence on measuring system time response and affects the output signal. This behavior can be studied effectively by applying a simple periodic waveform to the system. Consider the first-order measuring system to which an input of the form of a simple periodic function, $F(t) = A \sin \omega t$, is applied for $t \geq 0^+$:

$$\tau \dot{y} + y = KA \sin \omega t$$

with initial conditions $y(0) = y_0$. Note that ω in [rad/s] $= 2\pi f$ with f in [Hz]. The general solution to this differential equation yields the measurement system output signal, that is, the time response to the applied input, $y(t)$:

$$y(t) = Ce^{-t/\tau} + \frac{KA}{\sqrt{1 + (\omega\tau)^2}} \sin\left(\omega t - \tan^{-1} \omega\tau\right) \tag{3.8}$$

where the value for C depends on the initial conditions.

So what has happened? The output signal, $y(t)$, of Equation 3.8 consists of a transient and a steady response. The first term on the right side is the transient response. As t increases, this term decays to zero and no longer influences the output signal. Transient response is important only during the initial period following the application of the new input. We already have information about the system transient response from the step function study, so we focus our attention on the second term, the steady response. This term persists for as long as the periodic input is maintained. From Equation 3.8, we see that the frequency of the steady response term remains the same as the input signal frequency, but note that the amplitude of the steady response depends on the value of the applied frequency, ω. Also, the phase angle of the periodic function has changed.

Equation 3.8 can be rewritten in a general form:

$$\begin{aligned} y(t) &= Ce^{-t/\tau} + B(\omega)\sin[\omega t + \Phi] \\ B(\omega) &= \frac{KA}{\sqrt{1 + (\omega\tau)^2}} \\ \Phi(\omega) &= -\tan^{-1}(\omega\tau) \end{aligned} \tag{3.9}$$

where $B(\omega)$ represents the amplitude of the steady response and the angle $\Phi(\omega)$ represents the *phase shift*. A relative illustration between the input signal and the system output response is given in Figure 3.11 for an arbitrary frequency and system time constant. From Equation 3.9, both B and ϕ are frequency dependent. Hence, the exact form of the output response depends on the value of the frequency

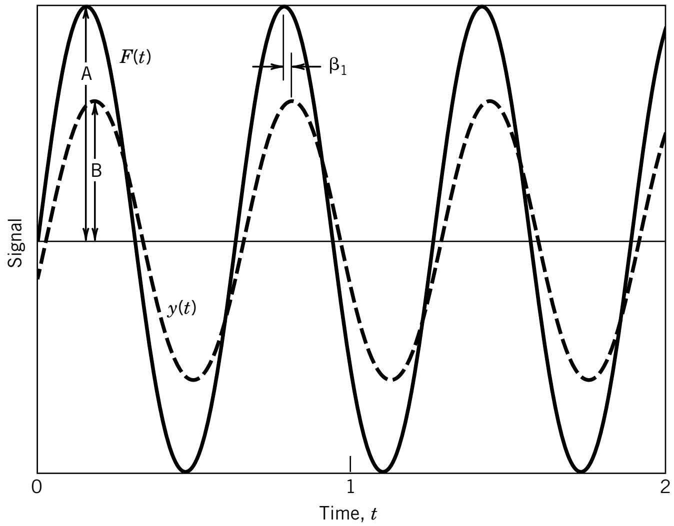


Figure 3.11 Relationship between a sinusoidal input and output: amplitude, frequency, and time delay.

of the input signal. The steady response of any system to which a periodic input of frequency, ω, is applied is known as the frequency response of the system. The frequency affects the magnitude of amplitude B and also can bring about a time delay. This time delay, β_1, is seen in the phase shift, $\Phi(\omega)$, of the steady response. For a phase shift given in radians, the time delay in units of time is

$$\beta_1 = \frac{\Phi}{\omega}$$

that is, we can write

$$\sin(\omega t + \Phi) = \sin\left[\omega\left(t + \frac{\Phi}{\omega}\right)\right] = \sin[\omega(t + \beta_1)]$$

The value for β_1 will be negative, indicating that the time shift is a delay between the output and input signals. Since Equation 3.9 applies to all first-order measuring systems, the magnitude and phase shift by which the output signal differs from the input signal are predictable.

We define a magnitude ratio, $M(\omega)$, as the ratio of the output signal amplitude to the input signal amplitude, $M(\omega) = B/KA$. For a first-order system subjected to a simple periodic input, the magnitude ratio is

$$M(\omega) = \frac{B}{KA} = \frac{1}{\sqrt{1 + (\omega\tau)^2}} \tag{3.10}$$

The magnitude ratio for a first-order system is plotted in Figure 3.12, and the corresponding phase shift is plotted in Figure 3.13. The effects of both system time constant and input signal frequency on frequency response are apparent in both figures. This behavior can be interpreted in the following manner. For those values of ωt for which the system responds with values of $M(\omega)$ near unity, the measurement system transfers all or nearly all of the input signal amplitude to the output and with very little time delay; that is, B will be nearly equal to KA in magnitude and $\Phi(\omega)$ will be near zero degrees. At large values of $\omega\tau$ the measurement system filters out any frequency information of the

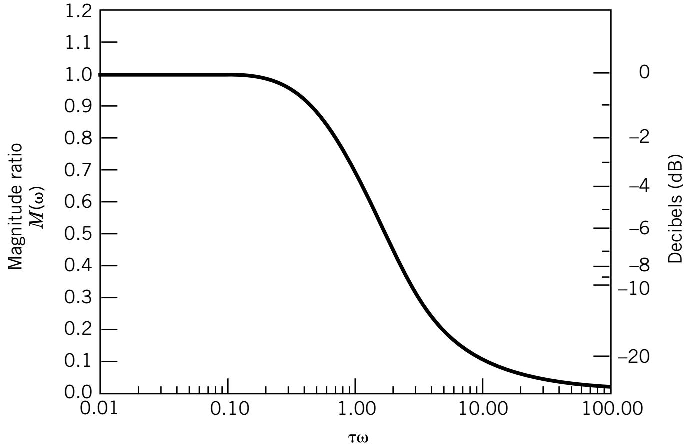


Figure 3.12 First-order system frequency response: magnitude ratio.

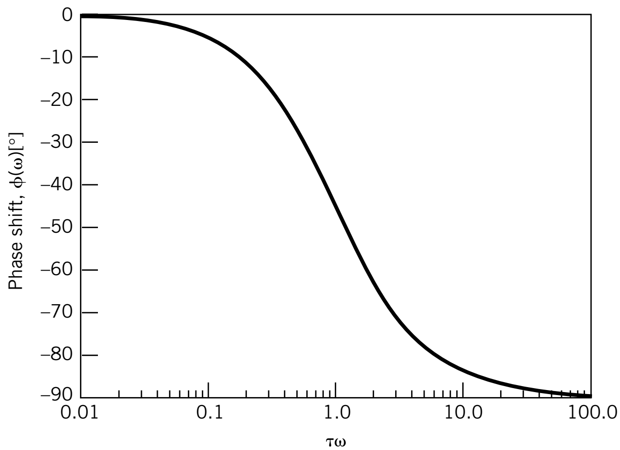


Figure 3.13 First-order system frequency response: phase shift.

input signal by responding with very small amplitudes, which is seen by the small $M(\omega)$, and by large time delays, as evidenced by increasingly nonzero β_1.

Any combination of ω and τ produces the same results. If we wanted to measure signals with high-frequency content, then we would need a system having a small τ. On the other hand, systems of large τ may be adequate to measure signals of low-frequency content. Often the trade-offs compete available technology against cost.

The *dynamic error*, $\delta(\omega)$, of a system is defined as $\delta(\omega) = M(\omega) - 1$. It is a measure of the inability of a system to adequately reconstruct the amplitude of the input signal for a particular input frequency. We normally want measurement systems to have a magnitude ratio at or near unity over the anticipated frequency band of the input signal to minimize $\delta(\omega)$. Perfect reproduction of the input signal is not possible, so some dynamic error is inevitable. We need some way to state this. For a first-order system, we define a *frequency bandwidth* as the frequency band over which $M(\omega) \geq 0.707$; in terms of the decibel (plotted in Figure 3.12) defined as

$$\mathrm{dB} = 20 \log M(\omega) \tag{3.11}$$

this is the band of frequencies within which $M(\omega)$ remains above -3 dB.

The functions $M(\omega)$ and $\Phi(\omega)$ represent the frequency response of the measurement system to periodic inputs. These equations and universal curves provide guidance in selecting measurement systems and system components.

Determination of Frequency Response The frequency response of a measurement system is found by a dynamic calibration. In this case, the calibration would entail applying a simple periodic waveform of known amplitude and frequency to the system sensor stage and measuring the corresponding output stage amplitude and phase shift. In practice, developing a method to produce a periodic input signal in the form of a physical variable may demand considerable ingenuity and effort. Hence, in many situations an engineer elects to rely on modeling to infer system frequency response behavior. We can predict the dynamic behavior if the time constant and static sensitivity of the system and the range of input frequencies are all known.

Example 3.6

A temperature sensor is to be selected to measure temperature within a reaction vessel. It is suspected that the temperature will behave as a simple periodic waveform with a frequency somewhere between 1 and 5 Hz. Sensors of several sizes are available, each with a known time constant. Based on time constant, select a suitable sensor, assuming that a dynamic error of $\pm 2\%$ is acceptable.

KNOWN $1 \le f \le 5$ Hz

$|\delta(\omega)| \le 0.02$

ASSUMPTIONS First-order system

$F(t) = A \sin \omega t$

FIND Time constant, τ

SOLUTION With $|\delta(\omega)| \le 0.02$, we would set the magnitude ratio between $0.98 \le M \le 1.02$. From Figure 3.12, we see that first-order systems never exceed $M = 1$. So the constraint becomes $0.98 \le M \le 1$. Then,

$$0.98 \le M(\omega) = \frac{1}{\sqrt{1 + (\omega\tau)^2}} \le 1$$

From Figure 3.12, this constraint is maintained over the range $0 \le \omega\tau \le 0.2$. We can also see in this figure that for a system of fixed time constant, the smallest value of $M(\omega)$ will occur at the largest frequency. So with $\omega = 2\pi f = 2\pi(5)$rad/s and solving for $M(\omega) = 0.98$ yields, $\tau \le 6.4$ ms. Accordingly, a sensor having a time constant of 6.4 ms or less will work.

Second-Order Systems

Systems possessing inertia contain a second-derivative term in their model equation (e.g., see Ex. 3.1). A system modeled by a second-order differential equation is called a second-order system. Examples of second-order instruments include accelerometers and pressure transducers (including microphones and loudspeakers).

In general, a second-order measurement system subjected to an arbitrary input, $F(t)$, can be described by an equation of the form

$$a_2\ddot{y} + a_1\dot{y} + a_0 y = F(t) \tag{3.12}$$

where a_0, a_1, and a_2 are physical parameters used to describe the system and $\ddot{y} = d^2y/dt^2$. This equation can be rewritten as

$$\frac{1}{\omega_n^2}\ddot{y} + \frac{2\zeta}{\omega_n}\dot{y} + y = KF(t) \tag{3.13}$$

where

$$\omega_n = \sqrt{\frac{a_0}{a_2}} = \text{natural frequency of the system}$$

$$\zeta = \frac{a_1}{2\sqrt{a_0 a_2}} = \text{damping ratio of the system}$$

Consider the homogeneous solution to Equation 3.13. Its form depends on the roots of the characteristic equation of Equation 3.13:

$$\frac{1}{\omega_n^2}\lambda^2 + \frac{2\zeta}{\omega_n}\lambda + 1 = 0$$

This quadratic equation has two roots,

$$\lambda_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

Depending on the value for ζ three forms of homogeneous solution are possible:

$0 \le \zeta < 1$ (underdamped system solution)

$$y_h(t) = Ce^{-\zeta\omega_n t}\sin\left(\omega_n\sqrt{1-\zeta^2 t} + \Theta\right) \tag{3.14a}$$

$\zeta = 1$ (critically damped system solution)

$$y_h(t) = C_1 e^{\lambda_1 t} + C_2 t e^{\lambda_2 t} \tag{3.14b}$$

$\zeta > 1$ (overdamped system solution)

$$y_h(t) = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t} \tag{3.14c}$$

The homogeneous solution determines the transient response of a system. The damping ratio, ζ, is a measure of system damping, a property of a system that enables it to dissipate energy internally. For systems with $0 \le \zeta \le 1$, the transient response will be oscillatory, whereas for $\zeta \ge 1$, the transient response will not oscillate. The critically damped solution, $\zeta = 1$, denotes the demarcation between oscillatory and nonoscillatory behavior in the transient response.

Step Function Input

Again, the step function input is applied to determine the general behavior and speed at which the system will respond to a change in input. The response of a second-order measurement system to a step function input is found from the solution of Equation 3.13, with $F(t) = AU(t)$, to be

$$y(t) = KA - KAe^{-\zeta\omega_n t}\left[\frac{\zeta}{\sqrt{1-\zeta^2}}\sin\left(\omega_n t\sqrt{1-\zeta^2}\right) + \cos\left(\omega_n t\sqrt{1-\zeta^2}\right)\right] \quad 0 \le \zeta < 1 \quad (3.15a)$$

$$y(t) = KA - KA(1+\omega_n t)e^{-\omega_n t} \quad \zeta = 1 \quad (3.15b)$$

$$y(t) = KA - KA\left[\frac{\zeta+\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}}e^{\left(-\zeta+\sqrt{\zeta^2-1}\right)\omega_n t} + \frac{\zeta-\sqrt{\zeta^2-1}}{2\sqrt{\zeta^2-1}}e^{\left(-\zeta-\sqrt{\zeta^2-1}\right)\omega_n t}\right] \quad \zeta > 1 \quad (3.15c)$$

where we set the initial conditions, $y(0) = \dot{y}(0) = 0$ for convenience.

Equations 3.15a–c are plotted in Figure 3.14 for several values of ζ. The interesting feature is the transient response. For under-damped systems, the transient response is oscillatory about the steady value and occurs with a period

$$T_d = \frac{2\pi}{\omega_d} = \frac{1}{f_d} \quad (3.16)$$

$$\omega_d = \omega_n\sqrt{1-\zeta^2} \quad (3.17)$$

where ω_d is called the *ringing frequency*. In instruments, this oscillatory behavior is called "ringing." The ringing phenomenon and the associated ringing frequency are properties of the measurement system and are independent of the input signal. It is the free oscillation frequency of a system displaced from its equilibrium.

The duration of the transient response is controlled by the $\zeta\omega_n$ term. In fact, its influence is similar to that of a time constant in a first-order system, such that we could define a second-order time constant as $\tau = 1/\zeta\omega_n$. The system settles to KA more quickly when it is designed with a larger

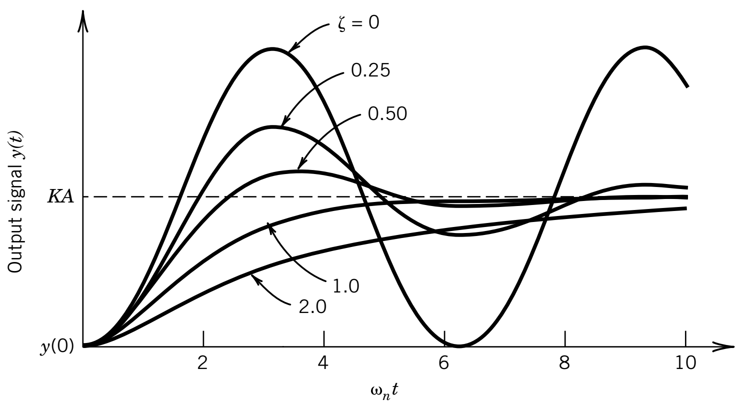


Figure 3.14 Second-order system time response to a step function input.

$\zeta\omega_n$ (i.e., smaller τ). Nevertheless, for all systems with $\zeta > 0$, the response eventually indicates the steady value of $y_\infty = KA$ as $t \to \infty$.

Recall that the *rise time* is defined as the time required to achieve a value within 90% of the step input. For a second-order system, the rise time is the time required to first achieve 90% of $(KA - y_0)$. Rise time is reduced by decreasing the damping ratio, as seen in Figure 3.14. However, the severe ringing associated with very lightly damped systems can delay the time to achieve a steady value compared to systems of higher damping. This is demonstrated by comparing the response at $\zeta = 0.25$ with the response at $\zeta = 1$ in Figure 3.14. With this in mind, the time required for a measurement system's oscillations to settle to within $\pm 10\%$ of the steady value, KA, is defined as its *settling time*. The settling time is an approximate measure of the time to achieve a steady response. A damping ratio of about 0.7 appears to offer a good compromise between ringing and settling time. If an error fraction (Γ) of a few percent is acceptable, then a system with $\zeta = 0.7$ will settle to steady response in about one-half the time of a system having $\zeta = 1$. For this reason, most measurement systems intended to measure sudden changes in input signal are typically designed such that parameters a_0, a_1, and a_2 provide a damping ratio of between 0.6 and 0.8.

Determination of Ringing Frequency and Rise and Settling Times The experimental determination of the ringing frequency associated with under-damped systems is performed by applying a step input to the second-order measurement system and recording the response with time. This type of calibration also yields information concerning the time to steady response of the system, which includes rise and settling times. Example 3.8 describes such a test. Typically, measurement systems suitable for dynamic signal measurements have specifications that include 90% rise time and settling time. *Adjust Second Order Parameters.vi* explores system values and response.

Determination of Natural Frequency and Damping Ratio From the under-damped system response to a step function test, values for the damping ratio and natural frequency can be extracted. From Figure 3.14, we see that with ringing the amplitude decays logarithmically with time towards a steady-state value. Let y_{max} represent the peak amplitude occurring with each cycle. Then for the first two successive peak amplitudes, let $y_1 = (y_{max})_1 - y_\infty$ and $y_2 = (y_{max})_2 - y_\infty$. The damping ratio is found from

$$\zeta = \frac{1}{\sqrt{1 + \left(2\pi / \ln\left({}^{y_1}/_{y_2}\right)\right)^2}} \tag{3.18}$$

From the calculation to find the ringing frequency using Equation 3.16, the natural frequency is found using Equation 3.17. Alternately, we could perform the step function test with $y(0) = KA$ and $y_\infty = 0$ and still use the same approach.

Example 3.7

Determine the physical parameters that affect the natural frequency and damping ratio of the accelerometer of Example 3.1.

KNOWN Accelerometer shown in Figure 3.3

ASSUMPTIONS Second-order system as modeled in Example 3.1

FIND ω_n, ζ

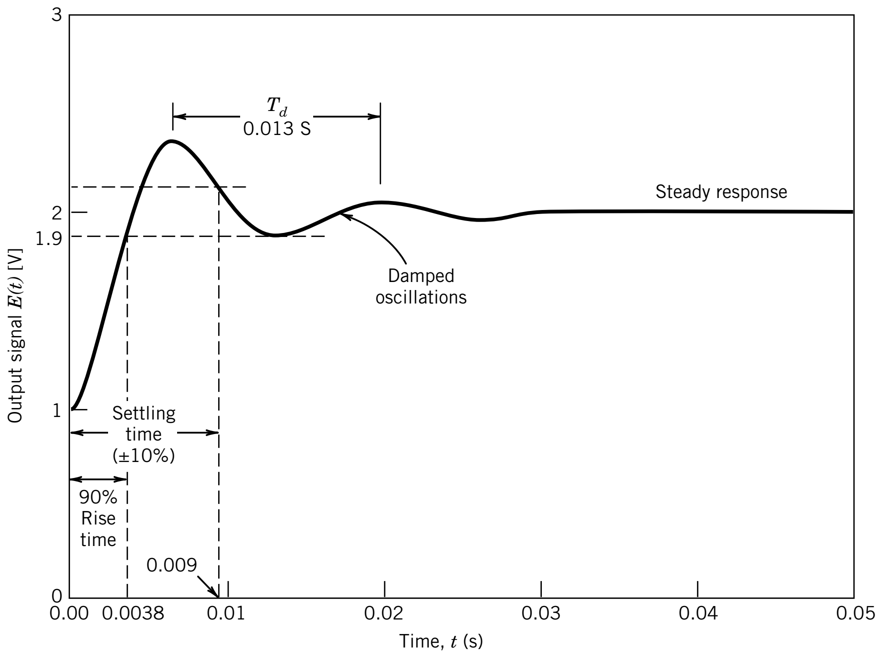


Figure 3.15 Pressure transducer time response to a step input for Example 3.8.

SOLUTION A comparison between the governing equation for the accelerometer in Example 3.1 and Equation 3.13 gives

$$\omega_n = \sqrt{\frac{k}{m}} \quad \zeta = \frac{c}{2\sqrt{km}}$$

Accordingly, the physical parameters of mass, spring stiffness, and frictional damping control the natural frequency and damping ratio of this measurement system.

Example 3.8

The curve shown in Figure 3.15 is the recorded voltage output signal of a diaphragm pressure transducer subjected to a step change in input. From a static calibration, the pressure–voltage relationship was found to be linear over the range 1 atmosphere (atm) to 4 atm with a static sensitivity of 1 V/atm. For the step test, the initial pressure was atmospheric pressure, p_a, and the final pressure was $2p_a$. Estimate the rise time, the settling time, and the ringing frequency of this measurement system.

KNOWN $p(0) = 1\text{ atm}$

$p_\infty = 2\text{ atm}$

$K = 1\text{ V/atm}$

ASSUMPTIONS Second-order system behavior

FIND Rise and settling times; ω_d

SOLUTION The ringing behavior of the system noted on the trace supports the assumption that the transducer can be described as having a second-order behavior. From the given information,

$$E(0) = Kp(0) = 1\text{ V}$$
$$E_\infty = Kp_\infty = 2\text{ V}$$

so that the step change observed on the trace should appear as a magnitude of 1 V. The 90% rise time occurs when the output first achieves a value of 1.9 V. The 90% settling time will occurs when the output settles between $1.9 < E(t) < 2.1$ V. From Figure 3.15, the rise occurs in about 4 ms and the settling time is about 9 ms. The period of the ringing behavior, T_d, is judged to be about 13 ms for an $\omega_d \approx 485$ rad/s.

Simple Periodic Function Input

The response of a second-order system to a simple periodic function input of the form $F(t) = A \sin \omega t$ is given by

$$y(t) = y_h + \frac{KA\sin[\omega t + \Phi(\omega)]}{\left\{\left[1 - (\omega/\omega_n)^2\right]^2 + [2\zeta\omega/\omega_n]^2\right\}^{1/2}} \tag{3.19}$$

with frequency-dependent phase shift

$$\Phi(\omega) = \tan^{-1}\left(-\frac{2\zeta\omega/\omega_n}{1 - (\omega/\omega_n)^2}\right) \tag{3.20}$$

The exact form for y_h is found from Equations 3.14a–c and depends on the value of ζ. The steady response, the second term on the right side, has the general form

$$y_{\text{steady}}(t) = y(t \to \infty) = B(\omega)\sin[\omega t + \Phi(\omega)] \tag{3.21}$$

with amplitude $B(\omega)$. Comparing Equations 3.19 and 3.21 shows that the amplitude of the steady response of a second-order system subjected to a sinusoidal input is also dependent on ω. So the amplitude of the output signal is frequency dependent. In general, we can define the magnitude ratio, $M(\omega)$, for a second-order system as

$$M(\omega) = \frac{B(\omega)}{KA} = \frac{1}{\left\{\left[1 - (\omega/\omega_n)^2\right]^2 + [2\zeta\omega/\omega_n]^2\right\}^{1/2}} \tag{3.22}$$

The magnitude ratio-frequency dependence for a second-order system is plotted in Figure 3.16 for several values of damping ratio. A corresponding plot of the phase-shift dependency on input frequency and damping ratio is shown in Figure 3.17. For an ideal measurement system, $M(\omega)$ would equal unity and $\Phi(\omega)$ would equal zero for all values of measured frequency. Instead, $M(\omega)$ approaches zero and $\Phi(\omega)$ approaches $-\pi$ as ω/ω_n becomes large. Keep in mind that ω_n is a property of the measurement system, while ω is a property of the input signal.

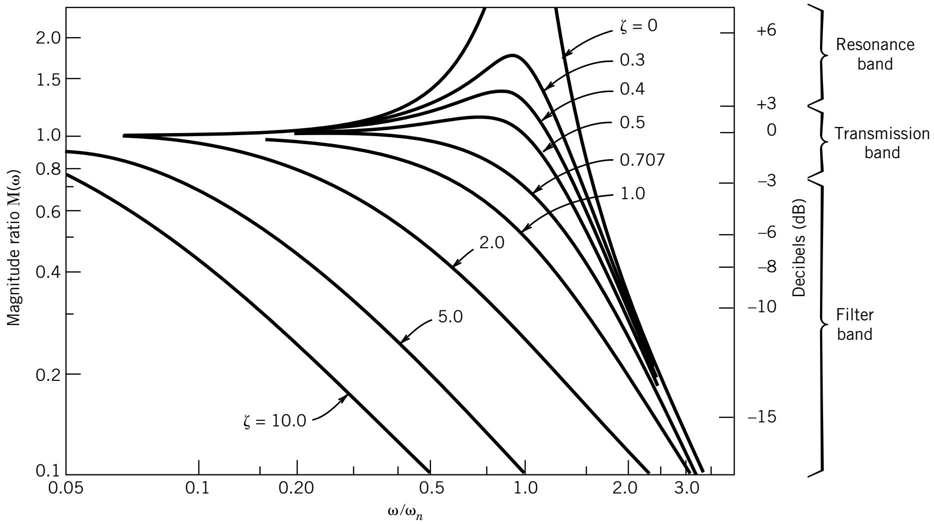


Figure 3.16 Second-order system frequency response: magnitude ratio.

System Characteristics

Several tendencies are apparent in Figures 3.16 and 3.17. For a system of zero damping, $\zeta = 0$, $M(\omega)$ will approach infinity and $\Phi(\omega)$ jumps to $-\pi$ in the vicinity of $\omega = \omega_n$. This behavior is characteristic of system resonance. Real systems possess some amount of damping, which modifies

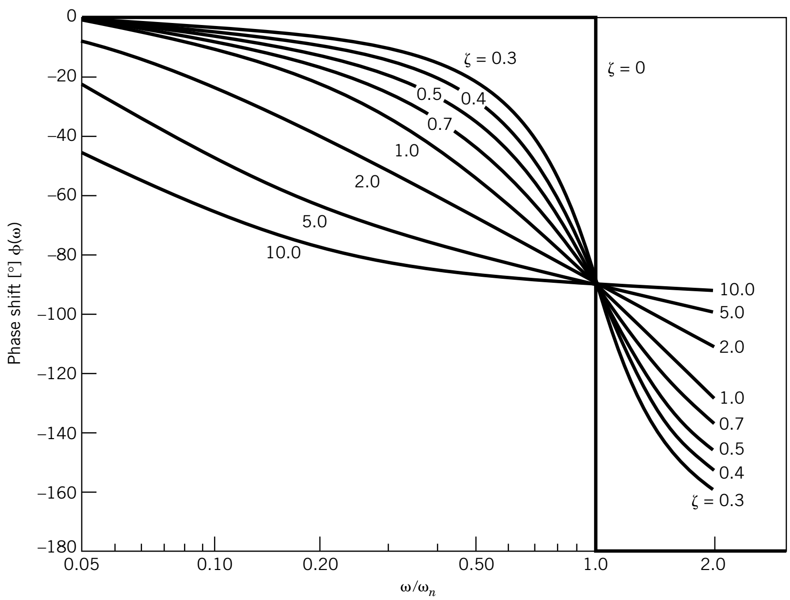


Figure 3.17 Second-order system frequency response: phase shift.

the abruptness and magnitude of resonance, but under-damped systems may still achieve resonance. This region on Figures 3.16 and 3.17 is called the *resonance band* of the system, referring to the range of frequencies over which the system is in resonance. Resonance in under-damped systems occurs at the *resonance frequency*,

$$\omega_R = \omega_n \sqrt{1 - 2\zeta^2} \tag{3.23}$$

The resonance frequency is a property of the measurement system. Resonance is excited by a periodic input signal frequency. The resonance frequency differs from the ringing frequency of free oscillation. Resonance behavior results in values of $M(\omega) > 1$ and considerable phase shift. For most applications, operating at frequencies within the resonance band is undesirable, confusing, and could even be damaging to some delicate sensors. Resonance behavior is very nonlinear and results in distortion of the signal. On the other hand, systems having $\zeta >$ 0.707 do not resonate.

At small values of ω/ω_n, $M(\omega)$ remains near unity and $\Phi(\omega)$ near zero. This means that information concerning the input signal of frequency ω will be passed through to the output signal with little alteration in the amplitude or phase shift. This region on the frequency response curves is called the *transmission band*. The actual extent of the frequency range for near unity gain depends on the system damping ratio. The transmission band of a system is either specified by its frequency bandwidth, typically defined for a second-order system as -3 dB $\leq M(\omega) \leq 3$ dB, or otherwise specified explicitly. You need to operate within the transmission band of a measurement system to measure correctly the dynamic content of the input signal.

At large values of ω/ω_n, $M(\omega)$ approaches zero. In this region, the measurement system attenuates the amplitude information in the input signal. A large phase shift occurs. This region is known as the *filter band*, typically defined as the frequency range over which $M(\omega) \leq -3$ dB. Most readers are familiar with the use of a filter to remove undesirable features from a desirable product. When you operate a measurement system within its filter band, the amplitudes of the portion of dynamic signal corresponding to those frequencies within the filter band will be reduced or eliminated completely. So you need to match carefully the measurement system characteristics with the signal being measured.

Example 3.9

Determine the frequency response of a pressure transducer that has a damping ratio of 0.5 and a ringing frequency (found by a step test) of 1200 Hz.

KNOWN $\zeta = 0.5$

$\omega_d = 2\pi(1200\,\text{Hz}) = 7540\,\text{rad/s}$

ASSUMPTIONS Second-order system behaviour

FIND $M(\omega)$ and $\Phi(\omega)$

SOLUTION The frequency response of a measurement system is determined by $M(\omega)$ and $\Phi(\omega)$ as defined in Equations 3.20 and 3.22. Since $\omega_d = \omega_n\sqrt{1 - \zeta^2}$, the natural frequency of the pressure transducer is found to be $\omega_n = 8706$ rad/s. The frequency response at selected frequencies is computed from Equations 3.20 and 3.22:

ω (rad/s)	$M(\omega)$	$\Phi(\omega)$ [°]
500	1.00	−3.3
2,600	1.04	−18.2
3,500	1.07	−25.6
6,155	1.15	−54.7
7,540	1.11	−73.9
8,706	1.00	−90.0
50,000	0.05	−170.2

COMMENT The resonance behavior in the transducer response peaks at $\omega_R = 6155$ rad/s. As a rule, resonance effects can be minimized by operating at input frequencies of less than ~30% of the system's natural frequency. The response of second-order systems can be studied in more detail using the companion software. Try the Matlab program *SecondOrd* and LabView program *Second_order*.

Example 3.10

An accelerometer is to be selected to measure a time-dependent motion. In particular, input signal frequencies below 100 Hz are of prime interest. Select a set of acceptable parameter specifications for the instrument assuming a dynamic error of ±5%.

KNOWN $f \leq$ 100Hz (i.e., $\omega \leq 628$ rad/s)

ASSUMPTIONS Second-order system
Dynamic error of ±5% acceptable

FIND Select ω_n and ζ

SOLUTION To meet a ±5% dynamic error constraint, we want $0.95 \leq M(\omega) \leq 1.05$ over the frequency range $0 \leq \omega \leq 628$ rad/s. This solution is open ended in that a number of instruments with different ω_n will do the task. So as one solution, let us set $\zeta = 0.7$ and then solve for the required ω_n using Equation 3.22:

$$0.95 \leq M(\omega) = \frac{1}{\left\{\left[1 - (\omega/\omega_n)^2\right]^2 + [2\zeta\omega/\omega_n]^2\right\}^{1/2}} \leq 1.05$$

With $\omega = 628$ rad/s, these equations give $\omega_n \geq 1047$ rad/s. We could plot Equation 3.22 with $\zeta = 0.7$, as shown in Figure 3.18. In Figure 3.18, we find that $0.95 \leq M(\omega) \leq 1.05$ over the frequency range $0 \leq \omega/\omega_n \leq 0.6$. Again, this makes $\omega_n \geq 1047$ rad/s acceptable. So as one solution, an instrument having $\zeta = 0.7$ and $\omega_n \geq 1047$ rad/s meets the problem constraints.

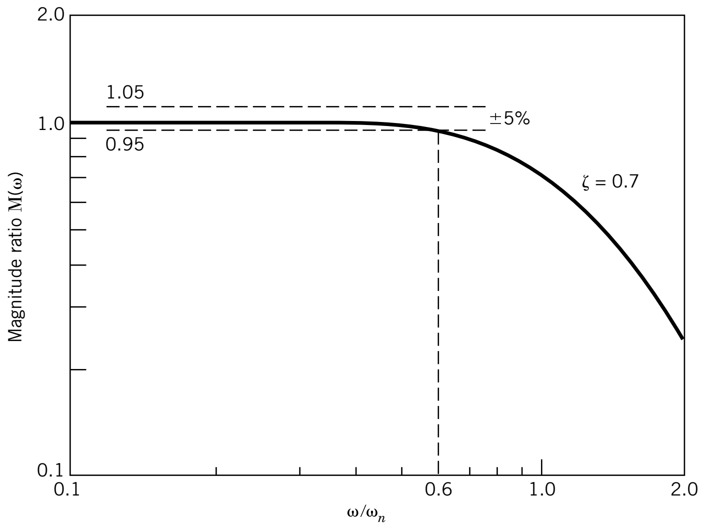


Figure 3.18 Magnitude ratio for second-order system with $\zeta = 0.7$ for Example 3.10.

3.4 TRANSFER FUNCTIONS

Consider the schematic representation in Figure 3.19. The measurement system operates on the input signal, $F(t)$, by some function, $G(s)$, so as to indicate the output signal, $y(t)$. This operation can be explored by taking the Laplace transform of both sides of Equation 3.4, which describes the general first-order measurement system. One obtains

$$Y(s) = \frac{1}{\tau s + 1} KF(s) + \frac{y_0}{\tau s + 1}$$

where $y_0 = y(0)$. This can be rewritten as

$$Y(s) = G(s)KF(s) + G(s)Q(s) \tag{3.24}$$

where $G(s)$ is the transfer function of the first-order system given by

$$G(s) = \frac{1}{\tau s + 1} \tag{3.25}$$

and $Q(s)$ is the system initial state function. Because it includes $KF(s)$, the first term on the right side of Equation 3.24 contains the information that describes the steady response of the measurement

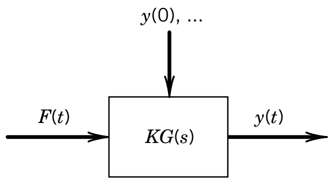


Figure 3.19 Operation of the transfer function. Compare with Figure 3.2.

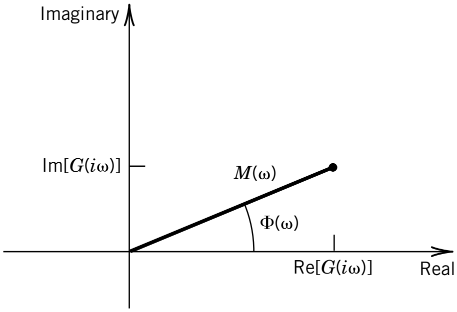


Figure 3.20 Complex plane approach to describing frequency response.

system to the input signal. The second term describes its transient response. Included in both terms, the transfer function $G(s)$ plays a role in the complete time response of the measurement system. As indicated in Figure 3.19, the *transfer function* defines the mathematical operation that the measurement system performs on the input signal $F(t)$ to yield the time response (output signal) of the system. As a review, Appendix C provides common Laplace transforms with their application to the solution of ordinary differential equations.

The system frequency response, which has been shown to be given by $M(\omega)$ and $\Phi(\omega)$, can be found by finding the value of $G(s)$ at $s = i\omega$. This yields the complex number

$$G(s = i\omega) = G(i\omega) = \frac{1}{\tau i\omega + 1} = M(\omega)e^{i\Phi(\omega)} \tag{3.26}$$

where $G(i\omega)$ is a vector on the real–imaginary plane having a magnitude, $M(\omega)$, and inclined at an angle, $\Phi(\omega)$, relative to the real axis as indicated in Figure 3.20. For the first-order system, the magnitude of $G(i\omega)$ is simply that given by $M(\omega)$ from Equation 3.10 and the phase shift angle by $\Phi(\omega)$ from Equation 3.9.

For a second-order or higher system, the approach is the same. The governing equation for a second-order system is defined by Equation 3.13, with initial conditions $y(0) = y_0$ and $\dot{y}(0) = \dot{y}_0$. The Laplace transform yields

$$Y(s) = \frac{1}{\left(1/\omega_n^2\right)s^2 + (2\zeta/\omega_n)s + 1}KF(s) + \frac{s\dot{y}_0 + y_0}{\left(1/\omega_n^2\right)s^2 + (2\zeta/\omega_n)s + 1} \tag{3.27}$$

which can again be represented by

$$Y(s) = G(s)KF(s) + G(s)Q(s) \tag{3.28}$$

By inspection, the transfer function is given by

$$G(s) = \frac{1}{\left(1/\omega_n^2\right)s^2 + (2\zeta/\omega_n)s + 1} \tag{3.29}$$

Solving for $G(s)$ at $s = i\omega$, we obtain for a second-order system

$$G(s = i\omega) = \frac{1}{(i\omega)^2/\omega_n^2 + 2\zeta i\omega/\omega_n + 1} = M(\omega)e^{i\Phi(\omega)} \tag{3.30}$$

which gives exactly the same magnitude ratio and phase shift relations as given by Equations 3.20 and 3.22.

3.5 PHASE LINEARITY

We can see from Figures 3.16 and 3.17 that systems having a damping ratio near 0.7 possess the broadest frequency range over which $M(\omega)$ will remain at or near unity and that over this same frequency range the phase shift essentially varies in a linear manner with frequency. Although it is not possible to design a measurement system without accepting some amount of phase shift, it is desirable to design a system such that the phase shift varies linearly with frequency. This is because a nonlinear phase shift is accompanied by a significant distortion in the waveform of the output signal. *Distortion* refers to a notable change in the shape of the waveform from the original, as opposed to simply an amplitude alteration or relative phase shift. To minimize distortion, many measurement systems are designed with $0.6 \leq \zeta \leq 0.8$.

Signal distortion can be illustrated by considering a particular complex waveform represented by a general function, $u(t)$:

$$u(t) = \sum_{n=1}^{\infty} \sin n\omega t = \sin \omega t + \sin 2\omega t + \cdots \tag{3.31}$$

Suppose during a measurement a phase shift of this signal were to occur such that the phase shift remained linearly proportional to the frequency; that is, the measured signal, $v(t)$, could be represented by

$$v(t) = \sin(\omega t - \Phi) + \sin(2\omega t - 2\Phi) + \cdots \tag{3.32}$$

Or, by setting

$$\theta = (\omega t - \Phi) \tag{3.33}$$

we write

$$v(t) = \sin\theta + \sin 2\theta + \cdots \tag{3.34}$$

We see that $v(t)$ in Equation 3.22 is equivalent to the original signal, $u(t)$. If the phase shift were not linearly related to the frequency, this would not be so. This is demonstrated in Example 3.11.

Example 3.11

Consider the effect of the variations in phase shift with frequency on a measured signal by examination of the signal defined by the function

$$u(t) = \sin t + \sin 5t$$

Suppose this signal is measured in such a way that a phase shift that is linearly proportional to the frequency occurs in the form

$$v(t) = \sin(t - 0.35) + \sin[5t - 5(0.35)]$$

Both $u(t)$ and $v(t)$ are plotted in Figure 3.21. We can see that the two waveforms are identical except that $v(t)$ lags $u(t)$ by some time increment.

Now suppose this signal is measured in such a way that the relation between phase shift and frequency was nonlinear, such as in the signal output form

$$w(t) = \sin(t - 0.35) + \sin(5t - 5)$$

The $w(t)$ signal is also plotted in Figure 3.21. It behaves differently from $u(t)$, and this difference is the signal distortion. In comparison of $u(t)$, $v(t)$, and $w(t)$, it is apparent that distortion is caused by a nonlinear relation between phase shift and frequency.

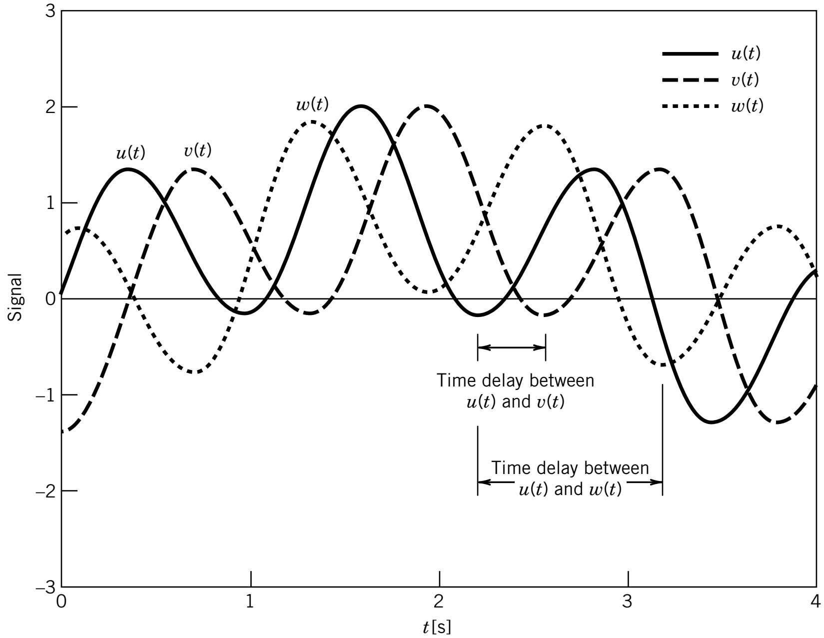


Figure 3.21 Waveforms for Example 3.11.

3.6 MULTIPLE-FUNCTION INPUTS

So far we have discussed measurement system response to a signal containing only a single frequency. What about measurement system response to multiple input frequencies? Or to an input that consists of both a static and a dynamic part, such as a periodic strain signal from an oscillating beam? When using models that are linear, such as ordinary differential equations subjected to inputs that are linear in terms of the dependent variable, the principle of superposition of linear systems applies in the solution of these equations. *The principle of superposition* states that a linear combination of input signals applied to a linear measurement system produces an output signal that is simply the linear addition of the separate output signals that would result if each input term had been applied separately. Because the form of the transient response is not affected by the input function, we can focus on the steady response. In general, we can write that if the forcing function of a form

$$F(t) = A_0 + \sum_{k=1}^{\infty} A_k \sin \omega_k t \tag{3.35}$$

is applied to a system, then the combined steady response will have the form

$$y_{\text{steady}}(t) = KA_0 + \sum_{k=1}^{\infty} B(\omega_k)\sin[\omega_k t + \Phi(\omega_k)] \tag{3.36}$$

where $B(\omega_k) = KA_k M(\omega_k)$. The development of the superposition principle can be found in basic texts on dynamic systems (4).

Example 3.12

Predict the steady output signal from a second-order instrument having $K = 1$ unit/unit, $\zeta = 2$, and $\omega_n = 628$ rad/s, which is used to measure the input signal

$$F(t) = 5 + 10\sin 25t + 20\sin 400t$$

KNOWN Second-order system
$K = 1$ unit/unit; $\zeta = 2.0$; $\omega_n = 628$ rad/s
$F(t) = 5 + 10 \sin 25t + 20 \sin 400t$

ASSUMPTIONS Linear system (superposition holds)

FIND $y(t)$

SOLUTION Since $F(t)$ has a form consisting of a linear addition of multiple input functions, the steady response signal will have the form of Equation 3.33 of $y_{\text{steady}}(t) = KF(t)$ or

$$y(t) = 5K + 10KM(25\,\text{rad/s})\sin[25t + \Phi(25\,\text{rad/s})] + 20KM(400\,\text{rad/s})\sin[400t + \Phi(400\,\text{rad/s})]$$

Using Equations 3.20 and 3.22, or, alternatively, using Figures 3.16 and 3.17, with $\omega_n = 628$ rad/s and $\zeta = 2.0$, we calculate

$$M(25\,\text{rad/s}) = 0.99 \qquad \Phi(25\,\text{rad/s}) = -9.1^\circ$$
$$M(400\,\text{rad/s}) = 0.39 \qquad \Phi(400\,\text{rad/s}) = -77^\circ$$

So that the steady output signal will have the form

$$y(t) = 5 + 9.9\sin(25t - 9.1^\circ) + 7.8\sin(400t - 77^\circ)$$

The output signal is plotted against the input signal in Figure 3.22. The amplitude spectra for both the input signal and the output signal are shown in Figure 3.23. Spectra can also be generated by using the accompanying software programs *FunSpect* and *DataSpect*.

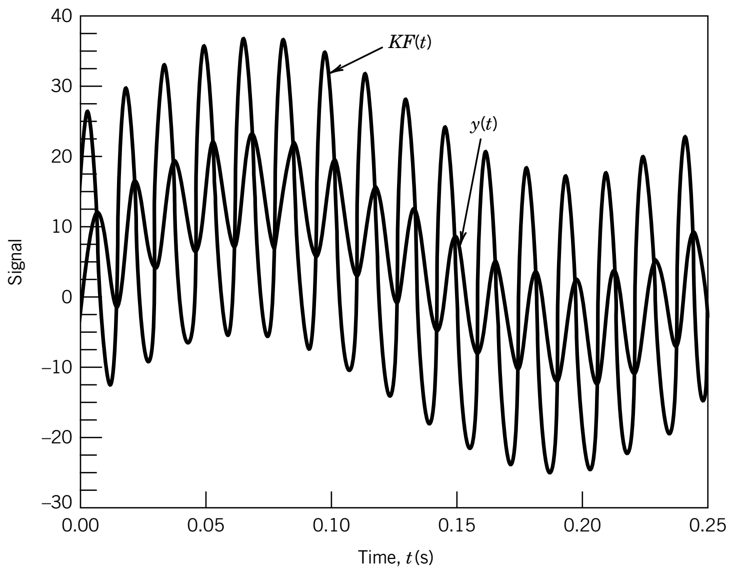


Figure 3.22 Input and output signals for Example 3.12.

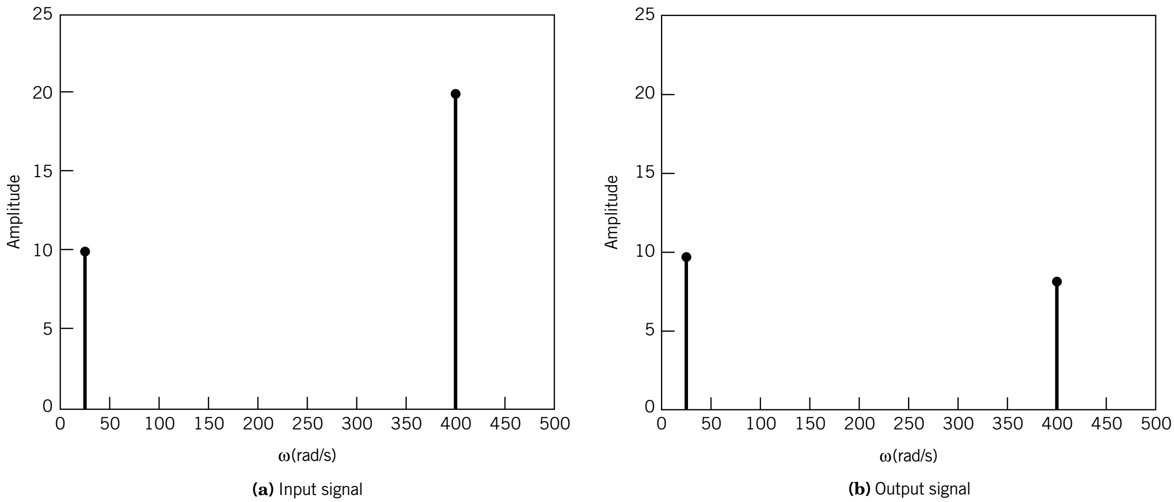


Figure 3.23 Amplitude spectrum of Example 3.12.

COMMENT The magnitude of average value and amplitude of the 25 rad/s component of the input signal are passed along to the output signal without much loss. However, the amplitude of the 400-rad/s component is severely reduced (down 61%). This is a filtering effect due to the measurement system frequency response.

3.7 COUPLED SYSTEMS

As instruments in each stage of a measurement system are connected (transducer, signal conditioner, output device, etc.), the output from one stage becomes the input to the next stage to which it is connected, and so forth. The overall measurement system will have a coupled output response to the original input signal that is a combination of each individual response to the input. However, the system concepts of zero-, first-, and second-order systems studied previously can still be used for a case-by-case study of the coupled measurement system.

This concept is easily illustrated by considering a first-order sensor that may be connected to a second-order output device (for example, a temperature sensor–transducer connected to a recorder). Suppose the input to the sensor is a simple periodic waveform, $F_1(t) = A \sin \omega t$. The transducer responds with an output signal of the form of Equation 3.8:

$$\begin{aligned} y_t(t) &= Ce^{-t/\tau} + \frac{K_t A}{\sqrt{1 + (\omega\tau)^2}} \sin(\omega t + \Phi_t) \\ \Phi_t &= -\tan^{-1} \omega\tau \end{aligned} \tag{3.37}$$

where the subscript t refers to the transducer. However, the transducer output signal now becomes the input signal, $F_2(t) = y_t$, to the second-order recorder device. The output from the second-order device, $y_s(t)$, will be a second-order response appropriate for input $F_2(t)$,

$$
\begin{aligned}
y_s(t) &= y_h(t) + \frac{K_t K_s A\sin[\omega t + \Phi_t + \Phi_s]}{\left[1 + (\omega\tau)^2\right]^{1/2} \left\{ \left[1 - (\omega/\omega_n)^2\right]^2 + [2\zeta\omega/\omega_n]^2 \right\}^{1/2}} \\
\Phi_s &= -\tan^{-1} \frac{2\zeta\omega/\omega_n}{1 - (\omega/\omega_n)^2}
\end{aligned}
\tag{3.38}
$$

where subscript s refers to the chart recorder and $y_h(t)$ is the transient response. The output signal displayed on the recorder, $y_s(t)$, is the measurement system response to the original input signal to the transducer, $F_1(t) = A \sin \omega t$. The steady amplitude of the recorder output signal is the product of the static sensitivities and magnitude ratios of the first- and second-order systems. The phase shift is the sum of the phase shifts of the two systems.

Based on the concept behind Equation 3.38 we can make a general observation. Consider the schematic representation in Figure 3.24, which depicts a measurement system consisting of H interconnected devices, $j = 1, 2, \ldots, H$, each device described by a linear system model. The overall transfer function of the combined system, $G(s)$, is the product of the transfer functions of each of the individual devices, $G_j(s)$, such that

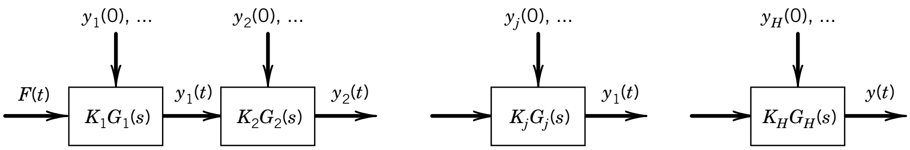
$$KG(s) = K_1 G_1(s) K_2 G_2(s) \ldots K_H G_H(s) \tag{3.39}$$

At $s = i\omega$, Equation 3.37 becomes

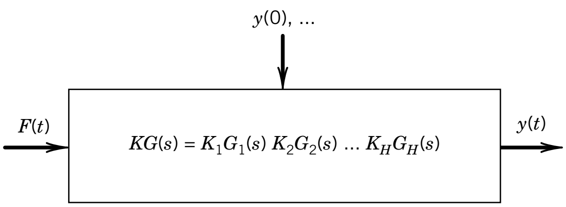
$$KG(i\omega) = (K_1 K_2 \ldots K_H) \times [M_1(\omega) M_2(\omega) \ldots M_H(\omega)] e^{i[\Phi_1(\omega) + \Phi_2(\omega) + \cdots + \Phi_H(\omega)]} \tag{3.40}$$

According to Equation 3.40, given an input signal to device 1, the system steady output signal at device H will be described by the system frequency response $G(i\omega) = M(\omega)e^{i\Phi(\omega)}$, with an overall system static sensitivity described by

$$K = K_1 K_2 \ldots K_H \tag{3.41}$$



(a) H-coupled transfer functions



(b) Equivalent system transfer function

Figure 3.24 Coupled systems: describing the system transfer function.

The overall system magnitude ratio is the product

$$M(\omega) = M_1(\omega)M_2(\omega)\ldots M_H(\omega) \tag{3.42}$$

and the overall system phase shift is the sum

$$\Phi(\omega) = \Phi_1(\omega) + \Phi_2(\omega) + \cdots + \Phi_H(\omega) \tag{3.43}$$

This holds true provided that significant loading effects do not exist, a situation discussed in Chapter 6.

3.8 SUMMARY

Just how a measurement system responds to a time-dependent input signal depends on the properties of the signal and the frequency response of the system. Modeling has enabled us to develop and to illustrate these concepts. Those system design parameters that affect system response are exposed through modeling, which assists in instrument selection. Modeling has also suggested the methods by which measurement system specifications such as time constant, response time, frequency response, damping ratio, and resonance frequency can be determined both analytically and experimentally. Interpretation of these system properties and their effect on system performance was determined.

The rate of response of a system to a change in input is estimated by use of the step function input. The system parameters of time constant, for first-order systems, and natural frequency and damping ratio, for second-order systems, are used as indicators of system response rate. The magnitude ratio and phase shift define the frequency response of any system and are found by an input of a periodic waveform to a system. Figures 3.12, 3.13, 3.16, and 3.17 are universal frequency response curves for first- and second-order systems. These curves can be found in most engineering and mathematical handbooks and can be applied to any first- or second-order system, as appropriate.

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NOMENCLATURE

$a_0, a_1, \ldots, a_n$	physical coefficients	m	mass (m)
$b_0, b_1, \ldots, b_m$	physical coefficients	$p(t)$	pressure $(m\, l^{-1}\, t^{-2})$
c	damping coefficient $(m\, t^{-1})$	t	time (t)
f	cyclical frequency $(f = \omega/2\pi)$ (Hz)	$x(t)$	independent variable
		$y(t)$	dependent variable
k	spring constant or stiffness $(m\, t^{-2})$	y^n	nth time derivative of $y(t)$

$y^n(0)$	initial condition of y^n	$U(t)$	unit step function
A	input signal amplitude	β_1	time lag (t)
$B(\omega)$	output signal amplitude	$\delta(\omega)$	dynamic error
C	constant	τ	time constant (t)
$E(t)$	voltage (V) or energy	$\Phi(\omega)$	phase shift
F	force $(m\,l\,t^{-2})$	ω	circular frequency (t^{-1})
$F(t)$	forcing function	ω_n	natural frequency magnitude(t^{-1})
$G(s)$	transfer function	ω_d	ringing frequency (t^{-1})
K	static sensitivity	ω_R	resonance frequency (t^{-1})
$M(\omega)$	magnitude ratio, B/KA	ζ	damping ratio
$T(t)$	temperature (°)	Γ	error fraction
T_d	ringing period (t)		

Subscripts

0	initial value
∞	final or steady value
h	homogeneous solution

PROBLEMS

Note: Although not required, the companion software can be used for solving many of these problems. We encourage the reader to explore the software provided.

3.1 A mass measurement system has a static sensitivity of 2 V/kg. An input range of 1 to 10 kg needs to be measured. A voltmeter is used to display the measurement. What range of voltmeter is needed. What would be the significance of changing the static sensitivity?

3.2 Determine the 75%, 90%, and 95% response time for each of the systems given (assume zero initial conditions):

a. $0.4\dot{T} + T = 4U(t)$

b. $\ddot{y} + 2\dot{y} + 4y = U(t)$

c. $2\ddot{P} + 8\dot{P} + 8P = 2U(t)$

d. $5\dot{y} + 5y = U(t)$

3.3 A special sensor is designed to sense the percent vapor present in a liquid–vapor mixture. If during a static calibration the sensor indicates 80 units when in contact with 100% liquid, 0 units with 100% vapor, and 40 units with a 50:50% mixture, determine the static sensitivity of the sensor.

3.4 A measurement system can be modeled by the equation

$$0.5\dot{y} + y = F(t)$$

Initially, the output signal is steady at 75 volts. The input signal is then suddenly increased to 100 volts.

a. Determine the response equation.

b. On the same graph, plot both the input signal and the system time response from $t = 0$ s through steady response.

3.5 Suppose a thermometer similar to that in Example 3.3 is known to have a time constant of 30 s in a particular application. Plot its time response to a step change from 32° to 120°F. Determine its 90% rise time.

3.6 Referring back to Example 3.3, a student establishes the time constant of a temperature sensor by first holding it immersed in hot water and then suddenly removing it and holding it immersed in cold water. Several other students perform the same test with similar sensors. Overall, their results are inconsistent, with estimated time constants differing by as much as a factor of 1.2. Offer suggestions as to why this might happen. Hint: Try this yourself and think about control of test conditions.

3.7 A thermocouple, which responds as a first-order instrument, has a time constant of 20 ms. Determine its 90% rise time.

3.8 During a step function calibration, a first-order instrument is exposed to a step change of 100 units. If after 1.2 s the instrument indicates 80 units, estimate the instrument time constant. Estimate the error in the indicated value after 1.5 s. $y(0) = 0$ units; $K = 1$ unit/unit.

3.9 Estimate any dynamic error that could result from measuring a 2-Hz periodic waveform using a first-order system having a time constant of 0.7 s.

3.10 A signal expected to be of the form $F(t) = 10 \sin 15.7t$ is to be measured with a first-order instrument having a time constant of 50 ms. Write the expected indicated steady response output signal. Is this instrument a good choice for this measurement? What is the expected time lag between input and output signal? Plot the output amplitude spectrum; $y(0) = 0$ and $K = 1$V/V.

3.11 A first-order instrument with a time constant of 2 s is to be used to measure a periodic input. If a dynamic error of $\pm 2\%$ can be tolerated, determine the maximum frequency of periodic input that can be measured. What is the associated time lag (in seconds) at this frequency?

3.12 Determine the frequency response [$M(\omega)$ and $\phi(\omega)$] for an instrument having a time constant of 10 ms. Estimate the instrument's usable frequency range to keep its dynamic error within 10%.

3.13 A temperature measuring device with a time constant of 0.15 s outputs a voltage that is linearly proportional to temperature. The device is used to measure an input signal of the form $T(t) = 115 + 12 \sin 2t$ °C. Plot the input signal and the predicted output signal with time assuming first-order behavior and a static sensitivity of 5 mV/°C. Determine the dynamic error and time lag in the steady response. $T(0) = 115$°C.

3.14 A first-order sensor is to be installed into a reactor vessel to monitor temperature. If a sudden rise in temperature greater than 100°C should occur, shutdown of the reactor will need to begin within 5 s after reaching 100°C. Determine the maximum allowable time constant for the sensor.

3.15 A single-loop *LR* circuit having a resistance of 1 MΩ is to be used as a low-pass filter between an input signal and a voltage measurement device. To attenuate undesirable frequencies above 1000 Hz by at least 50%, select a suitable inductor size. The time constant for this circuit is given by L/R.

3.16 A measuring system has a natural frequency of 0.5 rad/s, a damping ratio of 0.5, and a static sensitivity of 0.5 m/V. Estimate its 90% rise time and settling time if $F(t) = 2\,U(t)$ and the initial condition is zero. Plot the response $y(t)$ and indicate its transient and steady responses.

3.17 Plot the frequency response, based on Equations 3.20 and 3.22, for an instrument having a damping ratio of 0.6. Determine the frequency range over which the dynamic error remains within 5%. Repeat for a damping ratio of 0.9 and 2.0.

3.18 The output from a temperature system indicates a steady, time-varying signal having an amplitude that varies between 30° and 40°C with a single frequency of 10 Hz. Express the output signal as a waveform equation, $y(t)$. If the dynamic error is to be less than 1%, what must be the system's time constant?

3.19 A cantilever beam instrumented with strain gauges is used as a force scale. A step-test on the beam provides a measured damped oscillatory signal with time. If the signal behavior is second order, show how a data-reduction design plan could use the information in this signal to determine the natural frequency and damping ratio of the cantilever beam. (Hint: Consider the shape of the decay of the peak values in the oscillation.)

3.20 A step test of a transducer brings on a damped oscillation decaying to a steady value. If the period of oscillation is 0.577 ms, what is the transducer ringing frequency?

3.21 An input signal oscillates sinusoidally between 12 and 24 V with a frequency of 120 Hz. It is measured with an instrument with damping ratio of 0.7, ringing frequency of 1000 Hz, and static sensitivity of 1 V/V. Determine and plot the output signal amplitude spectrum at steady response.

3.22 An application demands that a sinusoidal pressure variation of 250 Hz be measured with no more than 2% dynamic error. In selecting a suitable pressure transducer from a vendor catalog, you note that a desirable line of transducers has a fixed natural frequency of 600 Hz but that you have a choice of transducer damping ratios of between 0.5 and 1.5 in. increments of 0.05. Select a suitable transducer.

3.23 A DVD/CD player is to be isolated from room vibrations by placing it on an isolation pad. The isolation pad can be considered as a board of mass m, a foam mat of stiffness k, and with a damping coefficient c. For expected vibrations in the frequency range of between 2 and 40 Hz, select reasonable values for m, k, and c such that the room vibrations are attenuated by at least 50%. Assume that the only degree of freedom is in the vertical direction.

3.24 A single-loop *RCL* electrical circuit can be modeled as a second-order system in terms of current. Show that the differential equation for such a circuit subjected to a forcing function potential $E(t)$ is given by

$$L\frac{d^2I}{dt^2} + R\frac{dI}{dt} + \frac{I}{C} = E(t)$$

Determine the natural frequency and damping ratio for this system. For a forcing potential, $E(t) = 1 + 0.5 \sin 2000t$ V, determine the system steady response when $L = 2$ H, $C = 1$ μF, and $R = 10{,}000$ Ω. Plot the steady output signal and input signal versus time. $I(0) = \dot{I}(0) = 0$.

3.25 A transducer that behaves as a second-order instrument has a damping ratio of 0.7 and a natural frequency of 1000 Hz. It is to be used to measure a signal containing frequencies as large as 750 Hz. If a dynamic error of ±10% can be tolerated, show whether or not this transducer is a good choice.

3.26 A strain-gauge measurement system is mounted on an airplane wing to measure wing oscillation and strain during wind gusts. The strain system has a 90% rise time of 100 ms, a ringing frequency of 1200 Hz, and a damping ratio of 0.8. Estimate the dynamic error in measuring a 1-Hz oscillation. Also, estimate any time lag. Explain in words the meaning of this information.

3.27 An instrument having a resonance frequency of 1414 rad/s with a damping ratio of 0.5 is used to measure a signal of ~6000 Hz. Estimate the expected dynamic error and phase shift.

3.28 Select one set of appropriate values for damping ratio and natural frequency for a second-order instrument used to measure frequencies up to 100 rad/s with no more than ±10% dynamic error. A catalog offers models with damping ratios of 0.4, 1, and 2 and natural frequencies of 200 and 500 rad/s. Explain your reasoning.

3.29 A first-order measurement system with time constant of 25 ms and $K = 1$ V/N measures a signal of the form $F(t) = \sin 2\pi t + 0.8 \sin 6\pi t$ N. Determine the steady response (steady output signal) from the system. Plot the output signal amplitude spectrum. Discuss the transfer of information from the input to the output. Can the input signal be resolved based on the output?

3.30 Demonstrate for the second-order system ($\omega_n = 100$ rad/s, $\zeta = 0.4$) subjected to step function input, $U(t)$, that the damping ratio can be found from the logarithmic amplitude decay whether $y(0) = 0$ with $F(t) = KAU(t)$ or $y(0) = KA$ with $F(t) = -KAU(t)$. Use $K = 1$ mV/mV and $A = 600$ mV. Do this by solving for the expected time response and from this response determine the successive peak amplitudes to extract the values for the damping ratio and natural frequency.

3.31 A force transducer having a damping ratio of 0.5 and a natural frequency of 4000 Hz is available for use to measure a periodic signal of 2000 Hz. Show whether or not the transducer passes a $\pm 10\%$ dynamic error constraint. Estimate its resonance frequency.

3.32 An accelerometer, whose frequency response is defined by Equations 3.20 and 3.22, has a damping ratio of 0.4 and a natural frequency of 18,000 Hz. It is used to sense the relative displacement of a beam to which it is attached. If an impact to the beam imparts a vibration at 4500 Hz, estimate the dynamic error and phase shift in the accelerometer output. Estimate its resonance frequency.

3.33 Derive the equation form for the magnitude ratio and phase shift of the seismic accelerometer of Example 3.1. Does its frequency response differ from that predicted by Equations 3.20 and 3.22? For what type of measurement would you suppose this instrument would be best suited?

3.34 Suppose the pressure transducer of Example 3.9 had a damping ratio of 0.6. Plot its frequency response $M(\omega)$ and $\phi(\omega)$. At which frequency is $M(\omega)$ a maximum?

3.35 A pressure transducer is attached to a stiff-walled catheter. The catheter is filled with saline from a small balloon attached at its tip. The initial system pressure is 50 mm Hg. At $t = 0$ s, the balloon is popped, forcing a step function change in pressure from 50 to 0 mm Hg. The time-based signal is recorded and the ringing period determined to be 0.03 s. Find the natural frequency and damping ratio attributed to the pressure-catheter system; $K = 1$ mV/mm Hg.

3.36 The output stage of a first-order transducer is to be connected to a second-order display stage device. The transducer has a known time constant of 1.4 ms and static sensitivity of 2 V/°C while the display device has values of sensitivity, damping ratio, and natural frequency of 1 V/V, 0.9, and 5000 Hz, respectively. Determine the steady response of this measurement system to an input signal of the form, $T(t) = 10 + 50 \sin 628t$ °C.

3.37 The displacement of a solid body is to be monitored by a transducer (second-order system) with signal output displayed on a recorder (second-order system). The displacement is expected to vary sinusoidally between 2 and 5 mm at a rate of 85 Hz. Select appropriate design specifications for the measurement system for no more than 5% dynamic error (i.e., specify an acceptable range for natural frequency and damping ratio for each device).

3.38 The input signal

$$F(t) = 2 + \sin 15.7t + \sin 160t \,\mathrm{N}$$

is applied to a force measurement system. The system has a known $\omega_n = 100$ rad/s, $\zeta = 0.4$, and $K = 1$ V/N. Write the expected form of the steady output signal in volts. Plot the resulting amplitude spectrum.

3.39 A signal suspected to be of the nominal form

$$y(t) = 5\sin 1000t \text{ mV}$$

is input to a first-order instrument having a time constant of 100 ms and $K = 1$ V/V. It is then to be passed through a second-order amplifier having a $K = 100$ V/V, a natural frequency of 15,000 Hz, and a damping ratio of 0.8. What is the expected form of the output signal, $y(t)$? Estimate the dynamic error and phase lag in the output. Is this system a good choice here? If not, do you have any suggestions?

3.40 A typical modern DC audio amplifier has a frequency bandwidth of 0 to 20,000 Hz $\pm$ 1 dB. Explain the meaning of this specification and its relation to music reproduction.

3.41 The displacement of a rail vehicle chassis as it rolls down a track is measured using a transducer ($K = 10$ mV/mm, $\omega_n = 10\,000$ rad/s, $\zeta = 0.6$) and a recorder ($K = 1$ mm/mV, $\omega_n = 700$ rad/s, $\zeta = 0.7$). The resulting amplitude spectrum of the output signal consists of a spike of 90 mm at 2 Hz and a spike of 50 mm at 40 Hz. Are the measurement system specifications suitable for the displacement signal? (If not, suggest changes.) Estimate the actual displacement of the chassis. State any assumptions.

3.42 The amplitude spectrum of the time-varying displacement signal from a vibrating U-shaped tube is expected to have a spike at 85, 147, 220, and 452 Hz. Each spike is related to an important vibrational mode of the tube. Displacement transducers available for this application have a range of natural frequencies from 500 to 1000 Hz, with a fixed damping ratio of about 0.5. The transducer output is to be monitored on a spectrum analyzer that has a frequency bandwidth extending from 0.1 Hz to 250 kHz. Within the stated range of availability, select a suitable transducer for this measurement from the given range.

3.43 A sensor mounted to a cantilever beam indicates beam motion with time. When the beam is deflected and released (step test), the sensor signal indicates that the beam oscillates as an underdamped second-order system with a ringing frequency of 10 rad/s. The maximum displacement amplitudes are measured at three different times corresponding to the 1st, 16th, and 32nd cycles and found to be 17, 9, and 5 mV, respectively. Estimate the damping ratio and natural frequency of the beam based on this measured signal, $K = 1$ mm/mV.

3.44 Write a short essay on how system properties of static sensitivity, natural frequency, and damping ratio affect the output information from a measurement system. Be sure to discuss the relative importance of the transient and steady aspects of the resulting signal.

3.45 Burgess (5) reports that the damping ratio can be approximated from the system response to an impulse test by counting the number of cycles, n, required for the ringing amplitudes to fall to within 1% of steady state by $\zeta = (4.6/2\pi n)$. The estimate is further improved if n is a noninteger. Investigate the quality of this estimate for a second-order system subjected instead to a step function input. Discuss your findings.

3.46 The starting transient of a DC motor can be modeled as an RL circuit, with the resistor and inductor in series (Figure 3.25). Let $R = 4\ \Omega$, $L = 0.1$ H, and $E_i = 50$ V with $t(0) = 0$. Find the current draw with time for $t > 0^+$.

3.47 A camera flash light is driven by the energy stored in a capacitor. Suppose the flash operates off a 6-V battery. Determine the time required for the capacitor stored voltage to reach 90% of its maximum energy ($= \frac{1}{2} CE_B^2$). Model this as an RC circuit (Fig. 3.26). For the flash: $C = 1000\ \mu$F, $R = 1$ kΩ, and $E_c(0) = 0$.

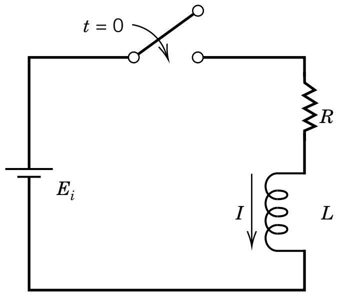

Figure 3.25 Figure for Problem 3.46.

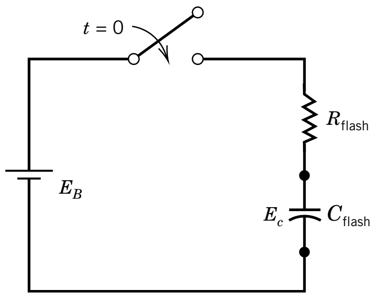

Figure 3.26 Figure for Problem 3.47.

3.48 Run program *Temperature Response*. The program allows the user to vary time constant τ and signal amplitude. To impose a step change move the input value up/down. Discuss the effect of time constant on the time required to achieve a steady signal following a new step change.

3.49 A lumped parameter model of the human systemic circulation includes the systemic vascular resistance and systemic vascular compliance. Compliance is the inverse of mechanical stiffness and is estimated by $C = \Delta\forall/\Delta p$, that is, the compliance is related to the pressure change required to accommodate a volume change. The typical stroke volume of each heart beat might eject 80 mL of blood into the artery during which the end-diastolic pressure of 80 mm Hg rises to a peak systolic pressure of 120 mm Hg. Similarly, the resistance to the 6 L/min of blood flow requires an average pressure over the pulse cycle of about 100 mm Hg to drive it. Estimate values for the systemic vascular compliance and resistance. Incidentally, these relations are similarly used in describing the physical properties of pressure measuring systems.